MILITARY SPECIFICATION

TEST STAND, JET ENGINE ACCESSORY OVERHAUL GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general design requirements for jet engine accessory overhaul hazardous and non-hazardous test stands for use in the overhaul depot.

1.2 Classification. The classification of test stands are as follows:

a. Hazardous test stands. Hazardous test stands are designed to comply with NFPA 70 Article 500, Class I, Division I, Group D, of the National Electrical Code. Hazardous test stands are test stands:

(1) In which hazardous concentrations of flammable gases or vapors are emitted continuously, intermittently or periodically under normal operation conditions. (See 6.3.1).

(2) In which hazardous concentrations of such gases or vapors may exist frequently because of repair or maintenance operations or because of leakage.

(3) In which breakdown or faulty operation of the test stand may release hazardous concentrations of flammable gases or vapors which might also cause simultaneous failure of the electrical equipment.

(4) Test stands which are not hazardous but will be utilized in hazardous locations. (See 6.2.b and 6.3.2).

b. Non-hazardous test stands. Non-hazardous test stands shall be test stands other than those described in 1.2.a.

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

W-F-406	Fittings for Cable, Power, Electrical and Conduit, Metal, Flexible
CC-M-636	Motors, Fractional Horsepower (Alternating - Current)
CC-M-641	Motor, Alternating Current (Integral Horsepower, 200 HP and Smaller)
CC-M-646	Motor, Direct Current (Integral Horsepower, 200 HP and Smaller)
WW-C-581	Conduit, Metal, Rigid, and Coupling, Elbow, and Nipple, Electrical Conduit, Zinc-Coated

Military

MIL-A-5498	Accumulators, Aircraft Hydropneumatic Pressure
MIL-H-5606	Hydraulic Fluid, Petrolum Base; Aircraft, Missile, and Ordinance
MIL-F-7024	Calibrating, Fluid, For Aircraft Fuel System Components
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base

STANDARDS

Federal

FED-STD-595	Colors.

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards, and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of

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MIL-STD-470	Maintainability Program Requirements (for Systems and Equipment)
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-831	Test Reports, Preparation of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-882	Systems Safety Program for Systems and Associated Subsystems and Equipment: Requirements for
MIL-STD-889	Dissimilar Metals
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MS 24380	Casters, Industrial
MS 28889	Valve, Air High Pressure Charging
MS 33656	Fitting End, Standard Dimensions For Flared Tube Connection and Gasket Seal

HANDBOOK

AFSC Design Handbook

DH 1-4	Electromagnetic Compatibility

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Fire Protection Association

NFPA 12	Carbon Dioxide Extinguishing Systems
NFPA 70	National Electrical Code
NFPA 493	Intrinsically Safe Process Control Equipment
NFPA 496	Purged Enclosures for Electrical Equipment

(Copies may be obtained from the National Fire Protective Association, 60 Batterymarch Street, Boston Massachusetts 02110.)

National Bureau of Standards

Handbook H28	Screw-Threads Standards for Federal Services

(Copies may be obtained from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

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American National Standards Institute

S1.4-1961 General Purpose Sound Level Meters
S1.11-1966 Octave, Half-Octave, and Third Octave Band Filter Sets

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10016.)

American Society of Mechanical Engineers

Unfired Pressure
Vessel Code

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 General specification. The individual item requirements shall be specified herein and in accordance with the applicable detail specifications.

3.2 Preproduction. This specification makes provisions for preproduction testing.

3.3 System safety and design review

3.3.1 System safety. The test stands shall be designed for system safety to comply with the Hazard Level, Category I - negligible conditions of MIL-STD-882 (see 6.3.3). A safety analyses shall be performed during the design of each type test stand to determine that the test stands will meet the safety requirements specified herein. Action shall be taken to eliminate or minimize hazards revealed by analyses. Unless otherwise specified (see 6.2.c), a safety analyses report shall be submitted and shall contain the information specified in Appendix IA.

3.3.2 Sound attenuation. Sound attenuation shall be provided as required in each type test stand to insure that the noise levels of the test stand, test component, and power conversion unit (see 3.10.11), during component testing, shall not exceed the limits specified in 3.3.2.1.

3.3.2.1 Noise levels. The noise levels produced by the test stand, test component and power conversion unit during component testing shall not exceed the sound pressure level requirements specified in table I, when measured as specified in 4.6.3.1 and figure 1.

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Table I Noise Level Requirements

Octave Band Center Frequencies (Hz)	Sound Pressure Levels (Decibels ref. 0.0002 dynes/cm^2) 10 ft from stand Operator's Position	
63	85	98
125	80	91
250	76	86
500	73	83
1000	71	81
2000	70	80
4000	69	80
8000	68	80

3.3.3 Belt drives. Belt drives shall not be used in hazardous test stands. Belt drives shall not be used in non-hazardous test stands except in self-contained variable speed drives.

3.3.4 Design reviews. Design reviews for each type test stand shall be conducted at the location designated by the procuring activity and design review data shall be submitted as specified herein. (See Appendix IB).

3.3.4.1 Preliminary design review. The preliminary design review shall consist of a joint Government and prime supplier review of the information specified in Appendix IB. The design review shall be held 90 days after Item Part I Specification approval (see 6.2.e).

3.3.4.2 Critical design review. The critical design review shall consist of a review of the changes/corrections made on the information presented at the preliminary design review. The critical design review shall be held a minimum of one week prior to ordering or parts and fabrication of the test stand (see 6.2.f).

3.3.5 Fire extinguishing system (H). Local application fire extinguishing system shall be provided on each test stand that tests fuel nozzles, nozzle clusters and spray rings. The system shall be of the automatic high pressure CO_2 system type in accordance with NFPA 12 with a NFPA approved Nationally recognized laboratory listed, spot type, combination rate of rise, 160°F fixed temperature detectors and nozzles located in the spray chambers and the cabinet enclosures.

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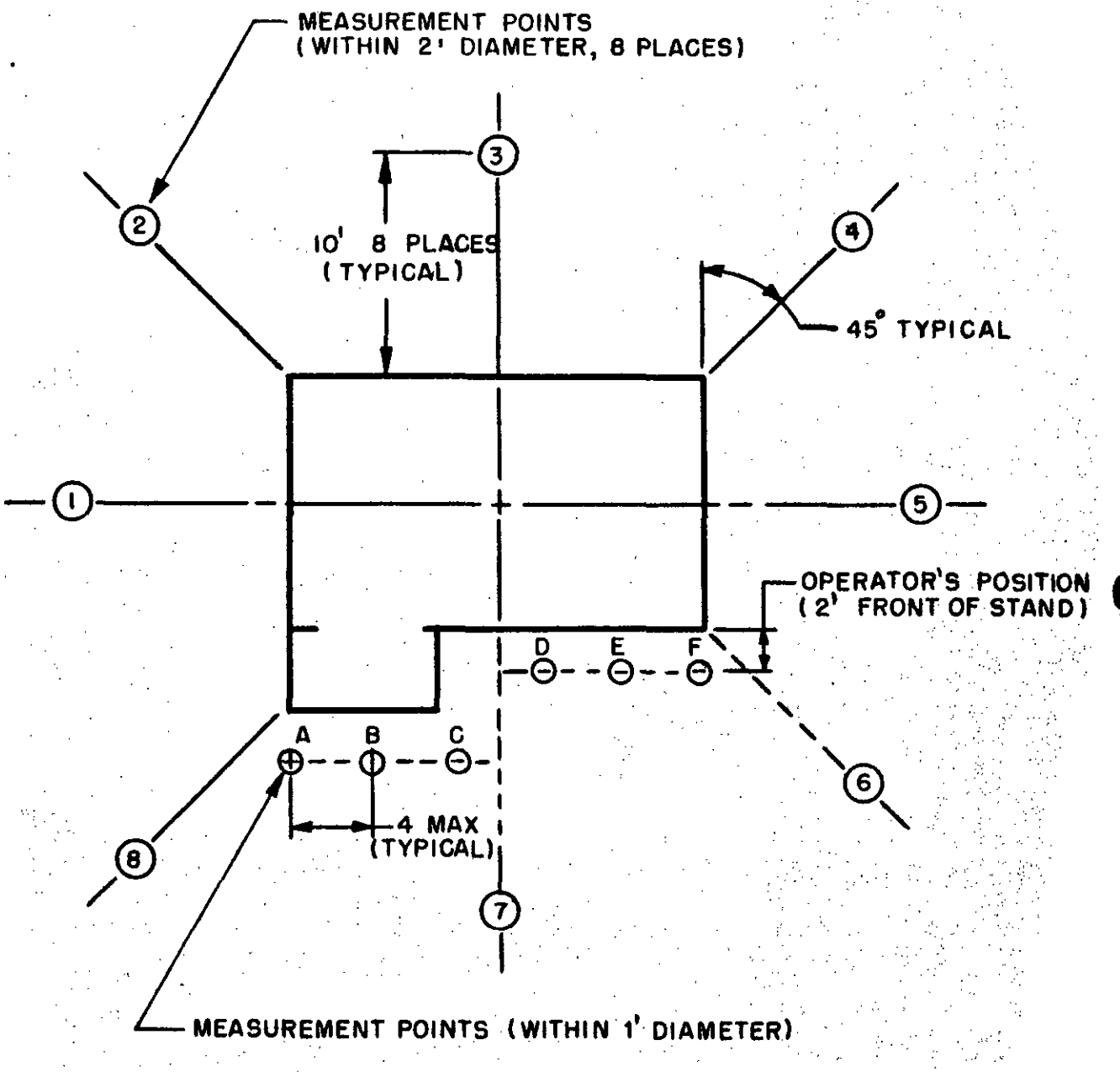


FIGURE 1. Noise Level Measurement Points

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3.3.5.1 Interlock (H). An interlock shall be provided on test stands with CO_2 system that upon manual or automatic release of the CO_2 system the emergency stop circuit is actuated.

3.3.5.2 Cylinder(s) location (H). Provisions shall be made for attaching a Government furnished CO_2 cylinder to the outside rear of the test stand.

3.3.6 Operator and maintenance personnel protection. All components and lines containing test fluid, air or steam exceeding 110°F shall be insulated to protect operating and maintenance personnel from burns. All control handles, knobs, and similar devices shall be maintained below 110°F. A tempered safety glass or equivalent shield shall be provided at the sink area to protect the test stand operator from an inadvertent leakage of hot test fluid that exceeds 110°F. A containment shield shall also be provided for protection from component failure during pressure testing (proof pressure test). (Examination, see 4.6.1).

3.4 Components. The test stand shall consist of the following major components as applicable to the test stand type.

Item or System Name	See Requirements
Fire extinguishing system	3.3.5
Test stand adapters	3.4.1
Panel layout	3.7.1.2
Cabinet	3.9
Air exhaust system	3.9.11
Electrical system	3.10
Hydraulic system	3.11
Sinks	3.11.11
Spray chambers	3.11.12
Reference orifices	3.11.13
Vacuum system	3.12
Pneumatic system	3.13
Water and steam system(s)	3.14
Instrumentation	3.15
Electrical/Electrical-mechanical instruments	3.15.1.4
Pressure measurement	3.15.2
Flow measurement	3.15.3
Speed measurement	3.15.4
Temperature measurement	3.15.5
Timers	3.15.6
Elapsed time meter	3.15.7
Timer	3.15.8
Hydrometer (H)	3.15.9
Accumulators	3.16

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3.4.1 Test stand adapters. Adapters that are required to mate test items to the test stand shall be provided with the test stand and be a component part of the test stand and so indicated on the test stand drawings. These adapters shall include such loose items as hose assemblies, test fixtures, cables, shields, supports etc. Special wrenches and hand tools that are required to teardown, assemble and adjust items such as fuel controls, pumps and similar items shall not be furnished as part of the test stand.

3.5 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.6 Materials. Materials used in the construction of the test stand shall be as specified herein. Materials shall not be hammered, filled, nor otherwise treated to conceal defects. Materials not definitely specified shall be of uniform quality and condition and free of seams, cracks, or other defects which may adversely affect the appearance, strength, endurance, or wear resistance of the finished test stand.

3.6.1 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.6.2 Protective treatment. Materials used in the construction of the test stands that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating and treatments shall be in accordance with MIL-STD-808.

3.6.3 Metals. Metals shall be of corrosion resistant materials, unless suitably protected to resist corrosion during normal service life and shall be suitable for the application intended.

3.6.3.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

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3.7 Design and construction

3.7.1 Design. The design, layout, and assembly of the test stand and its component parts shall be such as to facilitate the test operation and for ease of maintenance.

3.7.1.1 Grouping. All controls and displays which have sequential relations, associated with a particular function or operation, or are operated together shall be grouped together.

3.7.1.2 Test panel layout. The test panel layout shall be designed for operation by USAF 5th through 95 percentile Group as defined in MIL-STD-1472.

3.7.1.2.1 Instrumentation, controls and accessory drive location. All instrumentation, controls, and accessory drives shall be so located as to permit all phases of the test on the test component to be accomplished by one operator from the front of the test stand. Instrumentation shall be so located that the accuracy of the test will not be affected by improper readability or line of observation. All controls shall be selected and arranged to provide convenience and operating ease and to reduce test time and operator's fatigue with the ultimate purpose of obtaining maximum production from the test stand.

3.7.1.3 Service panel. A service panel shall be provided readily accessible through a cabinet access door which shall contain filter differential pressure gages, system bleed valves and other test stand set up and maintenance type component.

3.7.2 Reliability. The reliability of the test stand shall be controlled by the selection of test stand components (see Appendix IBe) that will provide satisfactory service life for production type testing. The design goal shall be a mean-time-between-failure of 1000 hours based on the confidence factor of 0.90. The test stand shall be designed for useful operating life of ten years.

3.7.3 Maintainability. The test stand shall be designed and constructed in accordance with the design guidelines of MIL-STD-470, paragraph titled Establish Maintainability Design Criteria. The guidelines shall be applied wherever appropriate to the subject class or type of test stand and wherever they do not conflict with other requirements specified herein. All maintenance, servicing, and adjustments shall be capable of being performed using only common handtools and equipment commonly expected in shops doing maintenance on similar equipment. (See 3.9.14).

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3.7.3.1 Maintenance downtime (see 6.3.5). The test stand shall be so designed and constructed to permit maintenance by Depot Test Stand military service technicians in not more than the downtimes listed below. Not more than two technicians shall be required simultaneously for any task.

3.7.3.1.1 Corrective maintenance downtime. The design mean corrective maintenance downtime shall be not more than 1 hour with a related maximum corrective maintenance downtime of not more than 3 hours.

3.7.3.1.2 Maximum scheduled maintenance downtime. The design maximum preventive maintenance downtime shall be not more than 16 manhours. This downtime shall include the time to service all required items and calibrate all indicating systems on the test stand.

3.7.4 Capability. The test stand shall be designed to test the jet engine accessory or component thereof within the operation and accuracy limits specified for that accessory or component.

3.7.4.1 Warmup time. Unless otherwise specified (see 6.2.i) (see 3.11.5) the test stand shall be capable of commencing the testing of the engine accessory or component 15 minutes after power is applied to the test stand.

3.7.5 Lubrication. All points of friction of the test stand controls and components shall be lubricated in accordance with MIL-STD-838 to ensure satisfactory operation.

3.7.5.1 Frequency of lubrication. The test stand shall be so designed that it will not require lubrication at intervals of less than 1 month when operated a minimum of 16 hours per day.

3.7.6 Plumbing. All hydraulic, pneumatic, steam and water lines, valves, fittings and similar components shall be selected and sized to permit safe operation under maximum pressure and flow conditions. Sizing of hydraulic plumbing components shall be as described herein in the paragraph entitled, Plumbing Lines, Fitting and Valves Sizing, (see 3.11.1). The allowable pressure drop through strainers, filters, heat exchangers and flow turbines shall be as required in 4.6.1.

3.7.6.1 Rigid lines and fittings. The use of rigid lines shall be held to a minimum. When used, such lines shall be of corrosion-resistant steel or aluminum alloy. Rigid lines, fittings, and components shall withstand 200 percent of the maximum rated pressure without leakage or permanent distortion. Commercial galvanized pipe and fittings will not be acceptable, except for water lines. Materials other than thread lubrication and polytetrafluoroethylene tape shall not be used for sealing threaded joints.

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3.7.6.1.1 Rigid line mounting. Rigid lines shall not be attached to the control panel mounted valves or fittings when sizes and pressures permit the use of flexible hose.

3.7.6.2 Flexible lines and fittings. Flexible AN or MS hose with AN- or MS- type fittings shall be used. Hose clamps shall not be used for fluid line couplings. All fluid lines, valves, and fittings shall be of a type rated by their manufacturers for the pressure and service to be encountered and shall withstand 200 percent of maximum operating pressure without leakage or permanent distortion.

3.7.6.2.1 Flexible hose. Flexible hose used in hazardous test stands shall be of the flexible steel braided type with AN-or MS- type fittings. The maximum resistance of the steel braid and end fittings assembly shall not exceed 100 ohms. The hose, fittings, and components shall withstand 200 percent of the maximum rated pressure without leakage or permanent distortion.

3.7.6.3 Threads

3.7.6.3.1 Screw threads. Screw threads shall conform to Handbook H-28.

3.7.6.3.2 Pipe threads. Pipe threads shall be avoided. Whenever pipe threads are used they shall be of standard commercial quality NPT. Permanent type pipe and gasket sealants shall not be used.

3.7.6.4 Service connections. Service connection for pressurized filling air, water and drains, etc., shall be located on the rear of the stand in horizontal or diagonal relative positions with a minimum distance of 6 inches between each connector with adequate space for flange fitting of 3 diameter lines or larger.

3.7.7 Foolproofness. Where improper installation of an item could cause malfunctioning of that item or system, in which it is installed, it shall be so designed that the item can only be installed in its proper operating position. (See 3.3, and 6.3.3).

3.7.8 Test stand components. All test stand components shall be new and of high quality and of current manufacture.

3.8 Performance. The test stand shall perform jet engine accessory or component acceptance and functional tests in accordance with applicable technical manuals and while being subjected to the environmental conditions

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3.8.1 Environmental conditions. The test stands shall be capable of operating under and satisfactorily resisting the following environmental conditions:

a. Hazardous test stands, operating temperatures ranging from +70 to +110°F.

b. Non-hazardous test stands, operating temperatures ranging from +45 to +125°F.

c. Humidity, 90 percent at 110°F.

3.9 Cabinet(s). Cabinet(s) shall be provided to house all components of the test stand including the remotely located components such as motor starters transformers and variable speed conversion units specified in 3.10.11. The cabinet(s) shall be all metal, fully enclosed type and suitably braced to carry the loads imposed during normal operation and transportation by common carrier. The cabinet(s) shall be fully guarded to protect operating personnel from fan blades, heating coils, electrical connections, and other components that may present a hazard environment. Test panels shall be constructed of 11-gage or heavier clear anodized aluminum. Doors and cabinet panels other than the test panels shall be constructed of lightweight and rigidly reinforced metal.

3.9.1 Cabinet size. Unless otherwise specified (see 6.2.j), the test stand cabinet shall be as small as possible to be consistent with good maintainability but not exceed maximum dimensions of 14 feet long, 7.5 feet wide, and 7.5 feet high. Centerline of read-out instruments shall not be mounted more than 6 feet above the facilities floor height (see 3.7.1.2.1). For test stands which are larger than these dimensions, the test stand shall be provided in sections, with each section having maximum dimensions of 14 feet long, 7.5 feet wide, and 7.5 feet high. Test panels that exceed 6 feet in width that do not have maintenance access doors on both sides shall have a maintenance walkway direct behind the test panel to allow easy access for maintenance on the rear of the test panel. The walkway(s) shall be a minimum of 2 feet wide and 5 feet in height. (See 3.9.13). Sharp objects projecting near the walkway shall be covered to prevent maintenance personnel injury.

3.9.1.1 Mating cabinets. Where sectional test stands are provided, all mating parts of each section, hydraulic, pneumatic, and electrical connections, shall be properly identified by permanent type tags and nameplates to insure and facilitate proper reassembly. Mating sections shall be mated in such a manner to prevent fluid leakage under the test stand. All sections shall be identified with the model and serial number of that stand.

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3.9.2 Handling provisions. Handling provisions shall be provided on the test stand in the form of forklift openings. One set of forklift openings shall be designed for lifting, supporting, and transporting the complete test stand, or each section if the test stand is provided in sections, with all equipment mounted therein.

3.9.2.1 Forklift openings. Two forklift openings shall be provided on the two opposite sides of the test stand or section. The centerline of each opening shall be a maximum of 27 ±1/2 inch from the center of gravity of the test stand or section and shall be equally spaced. Each opening shall be a minimum of 2.5 inches high, 9.75 inches wide, and capable of taking fork-lengths up to 80 inches long.

3.9.2.2 Casters. Casters shall be provided on portable test stands which require frequent movement. The casters shall conform to MS24380. Each test stand requiring casters shall be equipped with four swivel casters with a brake provided on each caster. Two of the casters shall be equipped with swivel locks to prevent caster swiveling while the stand is being pushed. Each caster used in the hazardous location shall be electrostatically conductive. Test stands incorporating casters shall include the handling provisions specified in 3.9.2 and 3.9.2.1.

3.9.3 Access doors. When required for maintenance purposes, access doors shall be provided on each side, rear, and front, except for the control panels, of the test stand. The access doors may provide a complete open area on each side, rear, and front except for the necessary bracing. If required for maintenance, the door bracing shall be removable. The access doors and mating parts shall be constructed to prevent inadvertent scraping of metal parts which might produce sparks.

3.9.3.1 Door size. Each door shall have a maximum width of 18 inches and shall extend to the full height of the test stand area whenever practical except for the necessary bracing. The access doors shall be provided in pairs wherever possible to provide a minimum free opening of 35 inches.

3.9.3.2 Door weight. Each door shall have a maximum weight of 38 pounds and so designed to be mounted on two hinge pins.

3.9.3.3 Door hinge pins. Each door shall be mounted on door hinge pins. The hinge pins shall be so designed to permit door removal by one man. The top hinge pin shall be of the retractable spring loaded type.

3.9.3.4 Hand latches. Positive flush-type hand latches shall be provided on each door.

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3.9.4 Top access. When required for maintenance purposes access through the top of the test stand shall be made through access panel(s). The panel(s) shall be retained in place with metal screws.

3.9.5 Working surface. A working surface shall be provided at the prime operator's position and at the component test position (see 6.3.6 and 6.3.7). The working surfaces shall be so designed and located to facilitate test component installation and removal and visual inspection of the test component from the operator's position. The working surfaces shall be of stainless steel. The design and construction of sinks and spray chambers are specified in 3.11.11 and 3.11.12.

3.9.5.1 Prime operator's position. The prime operator's position working surface shall be flat and free of obstructions for a minimum width of 18 inches and a minimum length of 18 inches. The working surface shall be 34 to 38 inches above floor level.

3.9.5.2 Component test position. The component test position working surface, top of sinks, etc., shall be 25 to 38 inches above floor level. Temperature simulator fluid wells shall be 44 to 48 inches above the floor level.

3.9.6 Toe clearance. Toe clearance shall be provided at the operator's position and component test position. The clearance shall be a minimum 6 inches in height and 4 inches in depth.

3.9.7 Handbook and record compartment. A compartment shall be provided for containing the equipment operation handbook and the equipment maintenance and inspection record. The compartment shall be a minimum size of 10 by 12 by 3 inches. This compartment shall be in a readily accessible location.

3.9.8 Equipment drawer. An equipment drawer shall be provided for storage of test stand components and hardware. The drawer shall be minimum size of 18 by 18 by 6 inches.

3.9.9 Vibration. The cabinet shall be so constructed that vibrations from the test stand components or component under test will not be transmitted through structural members to the instrumentation, thus impairing readability and accuracy.

3.9.10 Ventilation. Louvre-type air vents shall be provided in the cabinet near floor level and in the top to promote air circulation or cabinet shall be provided with forced ventilation.

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3.9.11 Air exhaust system. An air exhaust system shall be provided on test stands utilizing MIL-C-7024, Type II, MIL-L-7808, and MIL-H-5606, and high temperature silicon test fluids that incorporate sinks that may be wetted with these test fluids. The inlet to the air exhaust system shall be 3/8 ±1/8 inch-wide slot extending horizontally across the width of the sink. The inlet to the air exhaust system shall be on the test panel side of the sink and be even with or below the upper lip of the sink. The air exhaust system outlet shall terminate with a 4-inch diameter duct flush with the top of the test stand. The outlet and inlet of the air exhaust system within the test stand shall be connected with a 4-inch diameter duct and associated sheet metal.

3.9.12 Drip pan. A drip pan shall be provided in each test stand section that utilizes test fluids. The drip pan shall cover the leakage which may drip from any component part and flow down the inside walls. The drip pan shall be constructed of not less than 14-gage metal and shall have a minimum one-inch fall from each edge of the cabinet or section to the drain location.

3.9.12.1 Drain. The drip pan drain shall terminate at the rear of the test stand cabinet or each section if the test stand is provided in sections with a 1-inch female pipe fitting a minimum centerline distance of 4 inches above floor level. The drain shall be equipped with an NFPA approved Nationally recognized laboratory listed flame arrestor installed in an accessible location for cleaning.

3.9.13 Walkway(s). Walkway(s) shall be provided in each test stand or section as required, for test stand maintenance and inspection. The walkway(s) shall be of the open grate-type and located above the drip pan specified in 3.9.12.

3.9.13.1 Capacity and nonskid surface. Each walkway shall be capable of supporting a 400-pound load per square foot area and so designed or treated to provide a nonskid surface.

3.9.14 Maintenance hoisting provisions. Whenever the Government facilities hoisting provisions is inadequate for test stand maintenance (see 6.2.k), maintenance hoisting provisions shall be provided to remove components weighing:

a. Greater than 38 pounds and less than 76 pounds and two men cannot remove the component.

b. Greater than 76 pounds.

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3.10 Electrical system. The test stand electrical system shall be designed to supply power to the test stand components and the item under test, as required. Unless otherwise specified (see 6.2.1), the electrical system shall be designed to operate from a 277/480 volt, 3-phase, four wire, 60-cycle power source. (See 3.10.5). The electrical system shall consist of all electrically operated components within the test stand. NOTE: For test stands designed to operate in hazardous areas, the test stand electrical components shall be designed and installed in accordance with Article 500, NFPA 70, Class 1, Division 1, Group D location.

3.10.1 Sparking devices (H). All electrical sparking devices used in the hazardous test stands shall be provided in NFPA approved Nationally recognized laboratory listed explosion-proof enclosures (see 3.15.1.4).

3.10.2 Growth factor. The basic wiring of the test stand shall have a growth factor of not less than 15 percent to allow for additional instrumentation and control circuits.

3.10.3 Grounding. Test stands shall have all components electrostatically bonded together with an electrical continuity not to exceed 25 ohms resistance in accordance with NFPA 70, Article 250. A plainly marked grounding connection shall be provided on the test stand for connection to the facilities ground system.

3.10.4 Motors. All motors shall conform to CC-M-636, CC-M-641, or CC-M-646, as applicable. They shall be of adequate rating to operate under any conditions specified herein at maximum capacity without exceeding the NEMA recommended temperature limits for the class of insulating material used. The manufacturer's recommended full-load current shall not be exceeded during operation.

3.10.4.1 Drip-proof motors (N). Drip-proof motors shall be provided in non-hazardous hydraulic test stands.

3.10.4.2 Starters. Starters capable of withstanding openings under starting surges shall be provided. Motors producing 40 horsepower (hp) or more shall be provided with reduced voltage starters. The starters shall include overload relays for motor protection that can be manually reset without opening the starter case. (See 6.2.m).

3.10.4.2.1 Starter case(s). Starter case(s) shall be provided to enclose electric motor controls such as contactors, overload devices, transformers, starters, etc. The starter case(s) shall be so located for accessibility.

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3.10.4.2.1.1 Hinged covers (H). Hinged covers or double sections shall be provided for explosion-proof cases that are greater than 18 inches wide or high.

3.10.4.2.2 Overload protection. Overload protection shall be provided in all three phases or motors of 5 hp and larger.

3.10.4.3 Location (H). When specified (see 6.2.m), motors starters, relays, fuses, etc., for hazardous area test stands shall be located in a non-hazardous location area and site adapted to the location specified.

3.10.4.4 DC motors. DC motors shall be provided with a motor field loss relay which shall shut the drive down should the motor lose its field.

3.10.4.5 Interlocks. Where there is more than one motor in the test stand and possible damage may be caused by the stoppage of any one motor, interlocks shall be provided so that the stoppage of one motor in the circuit will not endanger or damage the test stand. Where the starting of the motors in an improper sequence can result in damage to the test stand, interlocks shall be provided to insure starting of all motors in proper sequence.

3.10.4.5.1 Automatic interlocks. Where air purging, cooling water or lubrication systems are required for test stand components, the controls for these systems shall be automatic or shall be interlocked with the motor controls in a manner to prevent damage to the test stand through operating error or an over-temperature or pressure condition.

3.10.5 Control, electrical instruments, and indicator light circuits. All controls, indicator lights, and electrical instruments, shall operate from 110 volt, single phase, 60-cycle power.

3.10.6 Vibration. All motors and motor pump sets shall be shock mounted to reduce vibration and noise to a minimum. The shock mounting shall effectively insulate the assembly from the test unit so as not to transmit any perceptible vibration to the installed instruments.

3.10.7 Wiring. All wiring between electrical components shall be in accordance with NFPA 70.

3.10.7.1 Sealing (H). For hazardous area test stands, fittings which require sealing in accordance with NFPA 70 for Class I, Division I, Group D hazardous locations shall be sealed (compounded) after the test stand has been installed in its Government (permanent) location. Fittings to be sealed shall be wrapped with a band of red tape. Tags attached to each fitting shall be marked Sealing Location. Crouse-Hinds Chico X fiber and Chico A sealing compound, or an equivalent approved by NFPA approved Nationally recognized laboratory listed

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sufficient for sealing these fittings shall be supplied and packed with each unit of the test stand. Circuits shall be checked for continuity, ground and correct hook-up, by reference to schematic(s) or manual covering the equipment. The wiring diagrams and drawings furnished with the test stand shall clearly indicate all fittings which will require sealing at the time of installation.

3.10.8 Cable conductor identification. Conductors shall be coded to conform with the code numbers of the wiring diagram. In shielded or jacketed conductors, the individual conductors shall be color-coded and the other jacket code numbered.

3.10.9 Cable and conduit fittings. Cable and conduit fittings shall conform to W-F-406 and installed in accordance with NFPA 70. Conduit shall be rigid zinc-coated steel conforming to WW-C-581 and shall be a minimum size of 3/4 inches.

3.10.10 Electrical connections. Electrical connections within the test stand shall be made by terminal strips, screw connections, and MS or AN connectors. The electrical power source connection shall be made at terminal strips in a junction box and located so the electrical power source wires shall enter the top of the test stand. A separate junction box shall be provided for lighting and control circuits.

3.10.10.1 Non-explosion proof connectors (H). Non-explosion proof connectors shall be permitted on hazardous test stands only at the component under test when required to mate with the component under test connector/receptacle.

3.10.10.1.1 Caution label. A red permanent type caution label shall be displayed near the non-explosion proof connectors indicating proper connection procedures.

3.10.11 Variable speed motor power conversion unit (H). When a variable speed motor power conversion unit is required for operation with hazardous area test stands, the unit shall be located in a non-hazardous location area and shall be site adapted to the location specified. (See 6.2.n).

3.10.11.1 Cabinet. When a variable speed motor power conversion unit is provided, the unit shall be housed in a cabinet. The cabinet shall meet the requirements of 3.9 through 3.9.4, 3.9.9 and 3.10.10.

3.10.11.2 Variable speed drive control. The variable speed drive control shall be designed so that the drive motor cannot be started unless the speed setting rheostat is set for minimum speed and shall provide infinitely variable adjustments throughout the speed range.

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3.10.11.3 Variable speed regulation. Variable speed regulation shall be ±0.1 percent of reading above 100 rpm. Variable speed regulation = $\frac{\text{(no load rpm} - \text{(full-load rpm)}}{\text{(no load rpm)}} \times 100$

3.10.11.3.1 Variable speed deviation. The maximum variable speed deviation from the set speed, upon sudden variation of load shall not exceed 1 percent of the set speed. All transients resulting from sudden load variations or sudden changes in the set speed shall be effectively damped out in less than 0.5 second.

3.10.11.3.2 Speed braking. Variable speed drives shall be equipped with a braking system as required to provide a speed reduction at a minimum rate of 300 rpm per second without having detrimental effect to the test stand or component under test.

3.10.11.3.2.1 Emergency speed braking. Emergency speed braking shall be provided for variable speed drives that shall bring the drive to a stop at a minimum rate of 300 rpm per second. This brake shall be activated by the emergency switch (see 3.10.14).

3.10.11.4 Safety interlocks. Safety interlocks shall be provided on all access doors on power conversion cabinets. The safety interlocks shall shut off power in the cabinet upon the opening of an access door.

3.10.12 DC power. When DC power is required for operation of the test stand or components thereof, the DC power supply shall be provided as a part of the test stand. The DC power supply shall consist of the necessary voltage regulating controls, ammeter(s), voltmeter(s), indicator light(s), switch(es), transformers, and other components as required to provide a complete power supply.

3.10.13 400-cycle power. When 400-cycle power is required for operation of the test stand or components thereof, the 400-cycle power supply shall be provided as a part of the test stand. The 400-cycle power supply shall consist of the necessary voltage and frequency regulating controls, ammeter(s), voltmeter(s), indicator light(s), switch(es), transformers, and other components as required to provide a complete power supply.

3.10.14 Emergency switch. A minimum of one emergency switch shall be provided on test stands (except non-hazardous electrical test stands) having a maximum length of 8 feet. A minimum of two emergency switches shall be provided on test stands greater than 8 feet in length. Each emergency switch shall be of the red, palm operated, mushroom-head, push-button-type and shall be located at the normal operator's position(s). Each emergency switch shall de-energize all electrical, air, water, hydraulic systems and

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activate the emergency braking circuit in the test stand. Primary consideration shall be personnel safety with provisions for protection of the test stand and component(s) under test added to the maximum extent practical.

3.10.15 Indicator lights. Green indicator lights shall be provided on the test stand to indicate when each electrical system circuit is activated and operating (such as motors, heaters, electric power on, etc.). Red indicator lights shall provide a warning or dangerous condition.

3.10.16 Maintenance lights. Interior lights shall be provided in the test stand to provide adequate lighting for maintenance purposes. A guard shall be provided around the light bulb to prevent personnel contact and inadvertent bulb breakage.

3.10.16.1 Interlock switch. The maintenance light circuit shall be interlocked with the service panel door so that the light will be off whenever the service panel door is closed.

3.10.17 Electromagnetic interference. Electrical and electronic components of the test stand shall not cause interference beyond the limits specified in MIL-STD-461 when tested in accordance with MIL-STD-462 test methods CE03 (Conducted Emission, 20 KHZ to 50 MHZ, power lead) and RE02 (Radiated Emission, 14 KHZ to 1 GHZ, Electric Field). When digital measuring devices and electrical process controllers are used with the test stands, test method RS02 (Radiated Susceptibility, Magnetic Induction Fields), shall apply. The AFSC Design Handbook 1-4 Electro-Magnetic Compatibility shall be used as a design guide toward achievement of specification compliance.

3.10.18 Annunicator. An annunicator shall be provided on test stands that have three or more automatic shut-down devices. The annunicator indicators shall be of the gravity drop-type. It shall be a NFPA approved Nationally recognized laboratory listed to be intrinsically safe in accordance with NFPA 493 or air purged in accordance with NFPA 496, type X.

3.11 Hydraulic system. When required by the test stand application, a hydraulic system shall be provided in the test stand. The hydraulic system shall include reservoir(s), pumps, filters, valves, etc., to provide a complete operational system for testing the test equipment. The hydraulic system shall operate satisfactorily with the reservoir filled to the 1/2-full position.

3.11.1 Plumbing lines, fittings and valve(s) sizing. Plumbing lines, fittings, valve(s) and other inline components in the supply line shall be sized in accordance with the flow requirements vs size and shall be equal or larger than that specified in table II. Plumbing lines, fittings, valve(s) and other inline components in the suction and return lines shall be sized in accordance with the flow requirements vs size and shall be equal or larger than that specified in table III.

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3.11.1.1 Gage port(s). Permanently installed and capped gage port(s) shall be provided at supply pump outlet(s), and test item inlet(s) when required, to allow the temporary installation of gage(s) to measure supply line pressure drops.

3.11.2 Reservoirs. Reservoirs shall be of welded corrosion resistant steel or aluminum alloy construction with the necessary inlets, outlets, valves, vents, and gage connections. Stainless steel, brass, or other non-corrosive fittings shall be attached by inert gas shielded arc-welding, brazing, silver-soldering, or other methods, as approved by the procuring activity. The method of attaching fittings shall be accomplished in a manner to minimize the possibility of electrolytic corrosion. All reservoirs shall be thoroughly neutralized and cleaned interiorly and exteriorly for removal of weld flux, scale, and other foreign matter. Rectangular reservoirs shall be reinforced and supported in such a manner to insure that the top of the reservoir will not sag.

3.11.2.1 Capacity. Unless otherwise specified, reservoirs shall have a minimum capacity in gallons equal to or greater than 1-1/2 times the maximum flow in gpm of the hydraulic system.

3.11.2.2 Pressure requirements. When reservoirs are required to maintain pressures less than atmospheric, the reservoir shall withstand a negative pressure of 0.2 psig without permanent distortion or leakage. Vented reservoirs provided in test stands shall withstand a positive pressure of 0.2 psig without permanent distortion or leakage. Positive pressurized reservoirs shall not be provided in the test stands.

3.11.2.3 Baffles. Baffles, if required, shall be provided in the reservoir to prevent pressure pulsations and minimize turbulence and aeration. The baffles shall provide clearance at the bottom of the reservoir to prevent an accumulation of foreign material.

3.11.2.4 Access plates. One or more access plates of at least 6-inch diameter in accordance with ASME Unfired Pressure Vessel Code shall be provided to permit visual inspection and complete access for manual cleaning of the interior of the reservoir.

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Table II. Component Supply Lines

AVERAGE VELOCITY = 15 FT/SEC

MIL-C-7024B TYPE II TEST FLUID PROPERTIES:
VISCOSITY @ 90 F = 1.119E-5 SQ.FT./SEC
DENSITY @ 90 F = 1.47 SLUGS/CU.FT

EQUIV MASS FLOW RATE (#\HR)	EQUIV VOL FLOW RATE (GPM)	INSIDE PIPE DIA (IN)	PRES DROP PER 25 FT (PSI)	POWER LOSS PER 25 FT (HP)	VALVE CV FACTORS
870.9	2.3	.2500	32.88	.044	.63
1360.7	3.6	.3125	24.81	.052	.99
1959.4	5.2	.3750	19.94	.060	1.42
2667.0	7.0	.4375	16.54	.068	1.93
3483.4	9.2	.5000	14.06	.075	2.53
4408.7	11.6	.5625	12.19	.083	3.20
5442.8	14.3	.6250	10.70	.089	3.95
6585.8	17.3	.6875	9.55	.097	4.77
7837.7	20.6	.7500	8.59	.103	5.68
9198.4	24.2	.8125	7.80	.110	6.67
10667.9	28.1	.8750	7.13	.117	7.73
12246.4	32.2	.9375	6.56	.123	8.88
13933.6	36.7	1.0000	6.07	.130	10.10
15729.9	41.4	1.0625	5.64	.136	11.40
17634.7	46.4	1.1250	5.27	.143	12.78
19648.6	51.7	1.1875	4.93	.149	14.24
21771.3	57.3	1.2500	4.63	.155	15.78
24002.9	63.2	1.3125	4.37	.161	17.40
26343.3	69.4	1.3750	4.14	.167	19.10
28792.5	75.8	1.4375	3.92	.173	20.87
31350.7	82.5	1.5000	3.72	.179	22.73
34017.7	89.6	1.5625	3.55	.186	24.66
36793.5	96.9	1.6250	3.38	.191	26.67
39678.2	104.5	1.6875	3.25	.198	28.76
42671.7	112.4	1.7500	3.09	.203	30.93
45774.1	120.5	1.8125	2.98	.209	33.18
48985.4	129.0	1.8750	2.85	.214	35.51
52305.5	137.7	1.9375	2.74	.220	37.92
55734.5	146.7	2.0000	2.64	.226	40.40
59272.4	156.1	2.0625	2.54	.231	42.97
62919.0	165.7	2.1250	2.46	.237	45.61
66674.6	175.6	2.1875	2.37	.243	48.33
70539.0	185.7	2.2500	2.29	.248	51.13

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Table II. Component Supply Lines (Cont'd)

AVERAGE VELOCITY = 15 FT/SEC

MIL-C-7024B TYPE II TEST FLUID PROPERTIES:
VISCOSITY @ 90 F = 1.119E-5 SQ.FT./SEC
DENSITY @ 90 F = 1.47 SLUGS/CU.FT.

EQUIV MASS FLOW RATE (#\HR)	EQUIV VOL FLOW RATE (GPM)	INSIDE PIPE DIA (IN)	PRES DROP PER 25 FT (PSI)	POWER LOSS PER 25 FT (HP)	VALVE CV FACTORS
74512.3	196.2	2.3125	2.22	.254	54.01
78594.4	206.9	2.3750	2.15	.260	56.97
82785.4	218.0	2.4375	2.09	.266	60.01
87085.2	229.3	2.5000	2.03	.271	63.13
91493.9	240.9	2.5625	1.97	.277	66.32
96011.4	252.8	2.6250	1.92	.283	69.60
100637.8	265.0	2.6875	1.86	.287	72.95
105373.1	277.4	2.7500	1.81	.293	76.39
110217.2	290.2	2.8125	1.76	.299	79.90
115170.2	303.2	2.8750	1.71	.303	83.49
120232.0	316.2	2.9375	1.67	.309	87.16
125402.7	330.2	3.0000	1.63	.313	90.91

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Table III. Component Return Lines

AVERAGE VELOCITY = 5 FT/SEC

MIL-C-7024B TYPE II TEST FLUID PROPERTIES:
VISCOSITY @ 90 F = 1.119E-5 SQ.FT./SEC
DENSITY @ 90 F = 1.47 SLUGS/CU.FT.

EQUIV MASS FLOW RATE (#\HR)	EQUIV VOL FLOW RATE (GPM)	INSIDE PIPE DIA (IN)	PRES DROP PER 25 FT (PSI)	POWER LOSS PER 25 FT (HP)	VALVE CV FACTORS
290.3	.8	.2500	4.93	.002	.47
453.6	1.2	.3125	3.73	.003	.74
653.1	1.7	.3750	2.97	.003	1.06
889.0	2.3	.4375	2.45	.003	1.44
1161.1	3.1	.5000	2.07	.004	1.88
1469.6	3.9	.5625	1.79	.004	2.38
1814.3	4.8	.6250	1.57	.004	2.94
2195.3	5.8	.6875	1.39	.005	3.56
2612.6	6.9	.7500	1.25	.005	4.23
3066.1	8.1	.8125	1.13	.005	4.97
3556.0	9.4	.8750	1.03	.006	5.76
4082.1	10.7	.9375	.94	.006	6.62
4644.5	12.2	1.0000	.87	.006	7.53
5243.3	13.8	1.0625	.81	.007	8.50
5878.2	15.5	1.1250	.75	.007	9.53
6549.5	17.2	1.1875	.70	.007	10.62
7257.1	19.1	1.2500	.66	.007	11.76
8001.0	21.1	1.3125	.62	.008	12.97
8781.1	23.1	1.3750	.59	.008	14.23
9597.5	25.3	1.4375	.55	.008	15.56
10450.2	27.5	1.5000	.52	.008	16.94
11339.2	29.9	1.5625	.50	.009	18.38
12264.5	32.3	1.6250	.47	.009	19.88
13226.1	34.8	1.6875	.45	.009	21.44
14223.9	37.5	1.7500	.43	.009	23.06
15258.0	40.2	1.8125	.41	.010	24.73
16328.5	43.0	1.8750	.40	.010	26.47
17435.2	45.9	1.9375	.38	.010	28.26
18578.2	48.9	2.0000	.37	.010	30.11
19757.5	52.0	2.0625	.35	.011	32.03
20973.0	55.2	2.1250	.34	.011	34.00
22224.9	58.5	2.1875	.33	.011	36.03
23513.0	61.9	2.2500	.32	.011	38.11
24837.4	65.4	2.3125	.31	.012	40.26
26198.1	69.0	2.3750	.30	.012	42.47
27595.1	72.7	2.4375	.29	.012	44.73

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Table III. Component Return Lines (Cont'd)

AVERAGE VELOCITY = 5 FT/SEC

MIL-C-7024B TYPE II TEST FLUID PROPERTIES:
VISCOSITY @ 90 F = 1.119E-5 SQ.FT./SEC
DENSITY @ 90 F = 1.47 SLUGS/CU.FT.

EQUIV MASS FLOW RATE (#\HR)	EQUIV VOL FLOW RATE (GPM)	INSIDE PIPE DIA (IN)	PRES DROP PER 25 FT (PSI)	POWER LOSS PER 25 FT (HP)	VALVE CV FACTORS
29028.4	76.4	2.5000	.28	.012	47.05
30498.0	80.3	2.5625	.27	.013	49.44
32003.8	84.3	2.6250	.26	.013	51.88
33545.9	88.3	2.6875	.25	.013	54.38
35124.4	92.5	2.7500	.25	.013	56.93
36739.1	96.7	2.8125	.24	.013	59.55
38390.1	101.1	2.8750	.23	.014	62.23
40077.3	105.5	2.9375	.23	.014	64.96
41800.9	110.1	3.0000	.22	.014	67.76
43560.7	114.7	3.0625	.22	.014	70.61
45356.9	119.4	3.1250	.21	.015	73.52
47189.3	124.2	3.1875	.20	.015	76.49
49058.0	129.2	3.2500	.20	.015	79.52
50963.0	134.2	3.3125	.19	.015	82.61
52904.2	139.3	3.3750	.19	.015	85.76
54881.8	144.5	3.4375	.19	.016	88.96
56895.7	149.8	3.5000	.18	.016	92.23
58945.8	155.2	3.5625	.18	.016	95.55
61032.2	160.7	3.6250	.17	.016	98.93
63154.9	166.3	3.6875	.17	.017	102.37
65313.9	172.0	3.7500	.17	.017	105.87
67509.2	177.8	3.8125	.16	.017	109.43
69740.7	183.6	3.8750	.16	.017	113.05
72008.6	189.6	3.9375	.16	.017	116.72
74312.7	195.7	4.0000	.15	.018	120.46
76653.1	201.8	4.0625	.15	.018	124.25
79029.8	208.1	4.1250	.15	.018	128.10
81442.8	214.4	4.1875	.15	.018	132.01
83892.1	220.9	4.2500	.14	.018	135.99
86377.6	227.4	4.3125	.14	.019	140.01
88899.5	234.1	4.3750	.14	.019	144.10
91457.6	240.8	4.4375	.14	.019	148.25
94052.0	247.6	4.5000	.13	.019	152.45
96682.7	254.6	4.5625	.13	.019	156.72
99349.7	261.6	4.6250	.13	.020	161.04
102053.7	268.7	4.6875	.13	.020	165.42

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Table III. Component Return Lines (Cont'd)

AVERAGE VELOCITY = 5 FT/SEC

MIL-C-7024B TYPE II TEST FLUID PROPERTIES:
VISCOSITY @ 90 F = 1.119E-5 SQ.FT./SEC
DENSITY @ 90 F = 1.47 SLUGS/CU.FT.

EQUIV MASS FLOW RATE (# HR)	EQUIV VOL FLOW RATE (GPM)	INSIDE PIPE DIA (IN)	PRES DROP PER 25 FT (PSI)	POWER LOSS PER 25 FT (HP)	VALVE CV FACTORS
104792.5	275.9	4.7500	.12	.020	169.86
107568.3	283.2	4.8125	.12	.020	174.36
110380.5	290.6	4.8750	.12	.020	178.92
113228.9	298.1	4.9375	.12	.021	183.54
116113.6	305.7	5.0000	.12	.021	188.21

3.11.2.5 Fluid level gage. A fluid level gage shall be provided on the reservoir. The gage shall be of the mechanical operated type. The gage indicator shall be marked at the 1/4-, 1/2-, 3/4-, and full positions and shall be red-lined at the 1/2-full position to indicate the minimum operating fluid level.

3.11.2.5.1 Accuracy. The fluid level gage accuracy shall be ±5.0 percent of reading.

3.11.2.5.2 Location. The fluid level gage shall be so located to permit observation during filling of the reservoir.

3.11.2.6 Fill connections. Manual and pressurized fluid filling connections shall be provided on each reservoir.

3.11.2.6.1 Manual filling connection. The manual filling connection on the reservoir shall be located inside the test stand cabinet and in a readily accessible position. The filling connection shall have a minimum 1-1/2 inch inside diameter opening and shall be provided with a NFPA approved Nationally recognized laboratory listed flame arrestor.

3.11.2.6.1.1 Fill cap. A sealed fill cap shall be provided for the manual filling connection. The cap shall be physically retained in the closed position.

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3.11.2.6.2 Pressurized filling. Pressurized filling shall be provided on each reservoir. The pressurized filling connection shall terminate flush at the rear cabinet surface with a 1-inch female pipe fitting.

3.11.2.6.2.1 Shutoff valve. A zero leakage shutoff valve shall be provided in the pressurized filling line and so located to permit the operator to operate the valve and observe the reservoir gage simultaneously.

3.11.2.6.2.2 Filter. A filter assembly capable of filtering 10-microns nominal shall be provided in the pressurized filling line and shall be equipped with AN-, MS- or commercial equivalent elements. The filter shall have a 10 gpm flow capacity at a pressure head of 10 psig.

3.11.2.7 Drainage. The reservoir shall be sloped to permit complete drainage of the fluid by gravity.

3.11.2.7.1 Drain line. A drain with a minimum inside diameter of 2 inches shall be provided. The drain line shall terminate with a standard female pipe coupling located at the rear of the test stand a minimum distance of 6 inches above the floor level. Zero leakage manual shutoff valves shall be provided in the drain line within the test stand.

3.11.2.7.1.1 Thermo-sensor (H). A thermo-sensor actuated valve normally closed shall be provided in the reservoir drain lines of hazardous test stands. Thermo-sensors shall be located above the test component and each test stand section. The sensors shall operate at not to exceed 160°F.

3.11.2.7.1.1.1 Interlock. An interlock shall be provided with the thermo actuated valve which shall cause actuation of the emergency stop circuit whenever the thermo-sensors operate.

3.11.2.8 Vents. Vent(s) on vented reservoirs shall terminate flush at the top of the test stand with a minimum 1-inch diameter female pipe fitting.

3.11.2.8.1 Vent flame arrestor (H). An NFPA approved Nationally recognized laboratory listed flame arrestor shall be installed between the vent line and the reservoir in hazardous test stands.

3.11.2.9 Suction and return ports. The suction and return ports shall be located as remotely as possible from each other. The return port shall be designed in such a manner that the return flow is directed away from the suction port and shall terminate 1-inch below the minimum operating fluid level.

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3.11.2.10 Overtemperature switch. The reservoir(s) shall be equipped with overtemperature switch(es) that will automatically shut the test stand off whenever the test fluid temperature exceeds 97 percent of the flash point of the fluid.

3.11.2.11 Low fluid level cutoff switch. A low fluid level switch shall be provided in the reservoir to shut the test stand down whenever the test fluid falls below the 1/2-full position.

3.11.3 Strainer. A 100-mesh, Y-Type, strainer shall be provided immediately downstream from the reservoir and so located for ease of installation and removal. The pressure drop through the strainer shall not exceed 5 psig.

3.11.3.1 Shutoff valve. A shutoff valve shall be provided between the strainer and reservoir to permit strainer maintenance.

3.11.4 Pressure pumps. Pressure pumps shall be provided as required for test stand operation. The pumps shall be of the direct drive type from the motors specified in 3.10.4, however, pumps mounted on the motor frame shall be so designed or equipped to prevent the leakage of a failing pump from entering the motor housing. Pumps shall be located for the greatest accessibility.

3.11.4.1 Low pressure pumps. Low pressure pumps shall be provided as required to boost the test fluid up to a maximum pressure of 100 psig. The shaft seal for pumps shall be a carbon seal rotating against steel, preferably with the steel surface stationary and the carbon seal rotating with the shaft. Seals shall be readily removable for replacement. Stuffing-box-type shaft seals shall not be used.

3.11.4.2 High pressure pumps. High pressure pumps shall be provided as required to boost test fluid above a pressure of 100 psig.

3.11.4.3 Pneumatic transportation. Pneumatics shall not be used as a method of transferring flammable or combustible fluids in test stands.

3.11.5 Filter assembly(ies). Filter assembly(ies) shall be provided downstream of each hydraulic pump. Each filter assembly shall withstand 200 percent of the maximum rated pressure of the test circuit without leakage or permanent distortion and shall be or corrosion resistant steel or aluminum alloy. Each filter assembly shall be equipped with a 10-micron nominal AN-, MS- or commercial equivalent, filter element(s), except on test stands utilizing MIL-L-7808 and MIL-H-5606 test fluid (see 3.11.5.1). The filter assembly shall be designed for flows at 200 percent the maximum rated flow capacity of the test circuit. Filter assemblies shall be so located to permit easy installation and removal of the filter element(s).

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3.11.5.1 Hydraulic oil filteration (N). Hydraulic filter assemblies on test stands utilizing MIL-L-7808 and MIL-H-5606 shall contain 5-micron absolute filter elements in the test item high pressure supply lines. Suction and return lines shall contain 10-micron nominal filter elements.

3.11.5.2 Filter drain. A filter assembly shall be provided with a drain to permit complete drainage of the fluid. A zero leakage hand valve shall be provided and located in an accessible location in the drain line.

3.11.5.3 Duplex pressure gage. A duplex pressure gage shall be provided on each filter assembly in lines of 100 psi and less to indicate the pressure drop across the filter. The pressure gage shall be located on the service panel.

3.11.5.3.1 Data tag. A permanent type data tag (see 3.22) shall be attached to the gage stating the maximum permissible pressure drop across the filter.

3.11.5.4 Differential pressure gage. A differential pressure gage shall be provided on each filter assembly in lines of 101 psi and over to indicate the pressure drop across the filter. The gage shall have a red-line above the maximum permissible pressure drop across the filter. The gage shall be located on the service panel.

3.11.5.5 Air purging. Provisions shall be provided for air purging filter assemblies that entrap air. The provisions shall consist of 1/8-inch return line to the reservoir with a shutoff valve and drip indicator in the line.

3.11.6 Test fluid heat exchangers. Heat exchangers for heating and cooling the test fluid, shall be provided when required. The pressure drop through the heat exchanger shall not exceed 10 psig.

3.11.6.1 Test fluid cooling below +80°F. When the test fluid must be maintained at the temperature of +80°F and below, the test fluid shall be cooled by mechanical refrigeration. For cooling between +80°F and -20°F, the refrigeration unit shall be a single stage type. For cooling from -20°F to -80°F, the refrigeration unit shall be of the cascade type. In test stands incorporating fluid wells, the test fluid cooling time shall not exceed a maximum of 45 minutes with a starting temperature of 70°F.

3.11.6.2 Test fluid cooling above +80°F. When the test fluid is at a temperature of +80°F and above and must be cooled, the test fluid shall be cooled with water at a temperature of 70°F maximum. When the test fluid is returned to the reservoir in hazardous area test stands, the heat exchanger shall be sized to provide an outlet test fluid temperature not to exceed 90 percent of flash point. Water flow shall not exceed 0.5 gpm per horsepower input to the hydraulic system and will be delivered at pressure ranging from 40 to 70 psig. With a water drain line back pressure of 4 psi or less.

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3.11.6.3 Test fluid heating. When the test fluid must be maintained at a temperature of +60° to +160°F, the test fluid shall be heated with water having a temperature of 180°F maximum. Water flow shall not exceed 0.5 gpm per horsepower input to the hydraulic system and will be delivered at pressures ranging from 40 to 70 psig. When the test fluid must be maintained at a temperature above 160°F, the test fluid shall be heated with steam having a maximum pressure of 15 psig. All components heated to 110°F and above shall be insulated to protect the operating and maintenance personnel from burns.

3.11.6.4 Immersion heaters. Immersion heaters shall not be used for heating MIL-F-7024 Type II test fluid. When immersion heaters are used in hydraulic fluid, silicon fluid and oils, the heaters shall be immersed in the fluid a minimum depth of 6 inches when the fluid is at its lowest operating level. The well or reservoir in which the immersion heaters are installed shall be equipped with an NFPA approved Nationally recognized laboratory listed automatic overtemperature shutoff switch that will automatically shut the heater off at 15°F above the maximum or reservoir operating temperature. The temperature of the fluid shall not exceed 80 percent of its auto-ignition temperature. The well or reservoir shall also be provided with a low fluid level switch that will shut down the test stand when the fluid level is less than 6 inches above the immersion heaters.

3.11.6.5 Fluid temperature controls. Automatic fluid temperature controls shall be provided for heating and cooling the test fluid. The control(s) shall maintain the test fluid at the required temperature to perform all tests on the test component with uninterrupted testing. The controls shall maintain a fluid temperature below 97 percent of the flashpoint of the test fluid except as specified in 3.11.6. Zero leakage shut off valves shall be provided on the inlet side of the water steam line.

3.11.6.5.1 In-line temperature-limiting switch(es). Unless otherwise specified, in-line temperature-limiting switches shall be provided in hazardous area test stands that automatically shut down the test stand when the in-line test fluid temperature reaches 97 percent of the test fluid flash point (see 3.11.7 and 3.11.6.5.1.1). These temperature-limiting switches shall be provided in all lines in which a temperature of 97 percent of the test fluid flash point can occur due to abnormal test stand operation or due to a component failure.

3.11.6.5.1.1 In-line hot test fluid temperature limiting switches. In-line hot test fluid (fluid temperature above 97 percent of the fluid flash point) switches shall be provided that automatically shut down the test stand when the test fluid rises to 15°F above the maximum test temperature required for the test component.

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3.11.6.6 Heat exchanger purging. Heat exchangers shall be equipped with purge systems when required to permit the release of air trapped in the cooling medium or in the test fluid.

3.11.7 Hot test fluid (H). Hot test fluid (test fluid temperature above 97 percent of the fluid flash point) shall be permitted only when required by the component to be tested. Test stand components used in the hot test fluid shall be kept to a minimum and shall be rated by the component manufacturer to satisfactorily operate at this temperature and pressure. The temperature of the test fluid shall not exceed 80 percent of its auto-ignition temperature.

3.11.7.1 Line length (H). The elevated temperature test fluid lines shall be of minimum length.

3.11.7.2 Steam heat exchanger (H). An insulated steam heat exchanger operating on 15 psi maximum steam pressure shall be provided for raising the test fluid temperature above 90°F and shall be functionally located immediately before the test component.

3.11.7.3 Water heat exchanger (H). A water heat exchanger shall be provided to lower the test fluid temperature to between 65°F and 110°F or 90 percent of the flash point of the test fluid (whichever is the lowest) before returning the fluid to the reservoir.

3.11.8 Open body tests (H). When an open body test is required on the test component and the test fluid is openly flowed into the sink or sump, the test fluid temperature shall not exceed 90 percent of the fluid's flash-point temperature.

3.11.9 Test fluid control valves. Test fluid control valves shall be provided as required to control the test fluid.

3.11.9.1 Control valves. Control valves shall be provided as required and shall be of the flow-type suitable for the intended application. Hysteresis shall be kept to a minimum and the valves shall not creep from selected positions.

3.11.9.1.1 Manual control valves. Manual control valves may be provided in fluid lines 1 inch and less in diameter. Where the control valve must be operated during the testing of the test component, the valve shall be mounted on the control panel. The valves shall so operate so that undue hydraulic shocks or surges shall not result from their use to change flow volume or direction. The maximum torque required to operate the valve shall not exceed 20 pound-inches of torque per inch of handle diameter. Push-pull valves requiring a maximum force of 7 pounds may be utilized.

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3.11.9.1.2 Air/electrical operated control valves. Air/electrical operated control valves shall be provided in fluid lines greater than 1 inch in diameter and may be provided in smaller fluid lines. The controls for air/electrical operated valves shall be mounted on the control console. The air/electrical operated valves shall return to the safe position in the event of control air pressure/electrical power loss and shall be capable of withstanding a continuous air pressure of 150 psig.

3.11.9.1.3 Automatic control valves. Automatic control valves shall be provided as required where the test fluid must be controlled to a constant pressure, flow or temperature.

3.11.9.1.4 Manual shut off valves. Manual zero leakage shut off valves shall be provided as required for maintenance and operation of the test stand. The maximum torque required to operate the valves shall not exceed 20 pound-inches of torque per inch of handle diameter.

3.11.9.1.5 Relief valves. Relief valves shall be provided as required to relieve excessive thermal expansion pressure and fluid pressure above operating pressure in the hydraulic system. The relief valves shall not chatter during operation.

3.11.10 Visible drip indicator. Visible drip indicators shall be provided on all test stand filter assemblies, heat exchangers, metered lines, and other locations as required for visible air purging and leakage indication that will affect satisfactory performance testing of the test components. The drain line from drip indicators shall terminate the reservoir.

3.11.10.1 Drip indicator(s) location. Drip indicators that will be observed during the operation of the test stand shall be located on the operator's panel. All other drip indicators shall be located on the service panel.

3.11.11 Sink(s). Sink(s) shall be provided to catch test fluid drainage leakage and external flow from fuel controls, governors and other test components.

3.10.11.1 Size. The size of the sink shall be a minimum of 2 feet in length and 18 inches in width. It shall be adequately sized to prevent splashing of test fluid outside of the sink and allow access for installing the test component.

3.11.11.2 Height. The height of the rim of the sink shall not exceed 38 inches above the floor. The bottom of the sink and the mounting of the test component in/or over the sink shall be so designed to position the component working area approximately 36 inches from the floor.

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3.11.11.3 Drains. Sink drain inlets shall be equipped with a NFPA approved Nationally recognized laboratory listed flame arrestor and an 80-mesh removable strainer. The drain shall terminate at the outside of the test stand with a female pipe coupling a minimum of 6 inches above the floor. The strainer shall be easily removable from the inlet of the drain. The sink drains for fuel controls, governors and similar type test components shall be equipped with a selector valve in the drain line that automatically directs the sink drainage to facilities drain whenever the test stand is shut-off. A manual selector shall be provided to select reservoir or facilities drain whenever the stand is running.

3.11.11.4 Materials. The material used in the fabrication of the inside and rim of the sink shall be number 4 standard polished (exposed side only) stainless steel.

3.11.12 Spray chambers (H). Spray chambers shall be provided for containing and collecting the flow, spray, and mist during the testing of nozzles, nozzle clusters, spray rings and similar type components.

3.11.12.1 Inside walls (H). The inside walls of the spray chamber shall be constructed of unpainted, unpolished, stainless steel and electro-conductive glass with a positive bond connection to the stand.

3.11.12.2 Observation window(s) size (H). Observation window(s) shall be provided. The size of each observation window shall be sufficient to allow adequate viewing of spray angle, skew angles, spray patterns and etc.

3.11.12.3 Blow-open exhaust panel(s) (H). Blow-open exhaust panel(s) or similar relief devices shall be provided on the spray chamber(s). Panel(s) shall be hinged on one side and be free to open in the event of an explosion in the chamber and shall open in a manner as not to endanger operating personnel. The panel(s) shall be so sized to relieve the gas pressures of an explosion without bursting the glass observation windows. A minimum of 1 square foot of opening per 15 cubic feet of chamber shall be provided.

3.11.12.4 Collector cans and spray deflectors (H). Stainless steel collector cans as required and metal wool-type spray deflectors on the floor of the spray chamber shall be provided to reduce the mist and aid test observation. These devices shall be electrical bonded to the test stand.

3.11.12.5 Test component height (H). The test component mounting height shall be so that an operator of 5 feet 5 inches to 6 feet 1 inch in height can view the test pattern in the standing position.

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3.11.12.6 Lighting (H). Lighting shall be provided in the spray chamber to satisfactorily observe the spray pattern, skew angle and etc.

3.11.12.7 Skew and angle measurement (H). Whenever required, skew, angle, and similar type measurement devices shall be provided in the spray chamber controlled from the outside of the spray chamber.

3.11.12.8 Mounting fixture rotation (H). The component mounting fixture shall be equipped with means of pneumatically rotating the test fixture in both directions to allow viewing of the complete pattern without the operator moving from the observation window.

3.11.12.9 Exhaust system (H). An exhaust system shall be provided with the spray chamber.

3.11.12.9.1 Exhaust blower (H). Exhaust blowers shall be provided with the spray chambers to remove all visible vapors from the spray chamber within 30 seconds after the conclusions of a test. The same blower shall be used to reduce visible vapors during the test.

3.11.12.9.2 Filter/Separator (H). A metallic-type electrical bonded air mist filter/separator shall be provided in the entrance to the exhaust blower. Filteration/separation shall be to the extent that fluid condensation shall not occur in the exhaust duct whenever the duct is at a temperature of 70°F and above.

3.11.12.9.3 Interlocks (H). The exhaust blower shall be electrical interlocked with the high pressure pump supply to the spray chamber so that the exhaust blower shall be running whenever fluid is supplied to the test chamber. The access door to the spray chamber shall be pneumatic/electrical interlocked to prevent the door from being opened while the high pressure supply pump is running. The spray chamber test fluid flow shall also be interlocked so that it can not be initiated until the chamber door is closed. No electrical components shall be installed in the spray chamber.

3.11.12.10 Drain (H). The spray chamber drain shall be directed to the stand reservoir.

3.11.12.10.1 Flame arrestor and strainer (H). A NFPA approved Nationally recognized laboratory listed flame arrestor and a 80-mesh removable strainer shall be provided in the inlet to the spray chamber drain.

3.11.13 Reference orifice(s) (H). Reference orifice(s) shall be provided with the first type of each fuel nozzle and fuel nozzle cluster test stands for determining the calibration of the test stand circuit.

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3.11.13.1 Calibration (H). Calibration of the orifice(s) shall be in accordance with the National Bureau of Standards (NBS) and the orifice(s) shall be stamped or tagged with the calibration data.

3.11.13.2 Size (H). The size of the orifice(s) opening shall be sized to produce a nominal test flow at nominal test pressure and temperature.

3.11.13.3 Circuit instrumentation accuracy (H). Flow, pressure and temperature instrumentation required in the orifice(s)/test component circuit(s) shall have sufficient accuracy to permit the performance of orifice flow type test to be run to determine if these instruments have shifted out of test calibration limits.

3.11.13.4 Limits (H). Calibration shift limits for the flow, pressure, and temperature instrumentation shall be provided with the orifice(s).

3.11.13.5 Calibration procedures (H). The calibration procedures for performing the orifice type test shall be with the temperature and pressure maintained at nominal, observe whether the flow is within limits.

3.11.13.6 Pressure fluctuation (H). Pressure fluctuation in the orifice shall not exceed plus or minus 0.10 psi and shall not oscillate at a rate greater than 0.1 psi per 2 seconds. The use of throttle valves and dampers in the pressure gage line shall not be permitted.

3.11.14 In-line orifice(s). In-line orifice(s) shall be provided as required. The housing of the orifice(s) shall be stamped or tagged with the orifice size.

3.11.15 Waste reservoir (N). A removable waste reservoir shall be provided in non-hazardous test stands that utilize MIL-L-7808 test fluid. The reservoir shall be located within the enclosure and have a minimum of 5 gallons capacity. The outlet of the waste reservoir when installed in the test stand shall be located a minimum of 16 inches above the floor and on the rear panel.

3.11.15.1 Shutoff valve (N). A shutoff valve shall be provided in the reservoir outlet.

3.11.16 Pressure and flow variation. Pressure and flow variation shall not vary more than 0.5 of the percent of the accuracy required of the pressure and flow instrumentation provided in the hydraulic circuit. Pressure variations can be dampened with the use of snubbers if the accuracy of the reading is not affected.

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3.12 Vacuum system. When a vacuum is required to operate the test stand or test a component, a complete vacuum system of sufficient capacity shall be provided. The vacuum exhaust for hazardous area test stands shall be vented to the air exhaust system, specified in 3.9.11, and for non-hazardous test stands shall be vented out the rear of the test stand and directed downward.

3.13 Pneumatic system. When pneumatics are required beyond shop availability to operate the test stand or test component, a complete pneumatic system of sufficient capacity and stability shall be provided (see 3.17).

3.13.1 Pressure vessels. Pressure vessels when required shall be provided in accordance with the ASME Unfired Pressure Vessel Code.

3.13.1.1 Pressure test date. The pressure and date when the pressure vessel was last hydrostatically tested shall be stenciled on the vessel.

3.13.2 Pneumatic supply. Where shop air at a pressure of 90 psig will satisfy the pneumatic supply requirements, provisions shall be provided on the test stand to mate with the shop air supply. Where high pressure pneumatics are required, necessary pressure boosting equipment shall be provided in the test stand to boost the shop air pressure.

3.13.2.1 Air loss automatic shutdown. An automatic operated pressure switch shall be provided in the pneumatic supply line which will automatically shut down the test stand safely upon air pressure loss.

3.13.2.2 Inlet air line(s). The inlet air line(s) to the test stand shall be equipped with pressure gage(s) relief valve(s) and 10-micron filter(s)/watertraps. The filter/watertrap shall be equipped with a differential pressure gage located on the service panel. All the other components shall be located immediately downstream of the air inlet to the test stand.

3.14 Water and steam systems. When water and steam is required in the operation of the test stand, the test stand shall contain all necessary water and steam plumbing and controls within the test stand. Provisions shall be provided on the test stand to mate with the facilities water and steam supply. A Y-type, 20-mesh strainer shall be provided in the immediate downstream of the water inlet to the test stand.

3.14.1 Water loss automatic shutdown. An automatic operated pressure switch shall be provided in the water supply line which will automatically shut down the test stand upon water pressure loss.

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3.15 Instrumentation. The test stand shall incorporate the necessary instrumentation systems for test stand operation and component testing except for the temperature simulations using capillary tubes required on fuel control test stands which shall be furnished as an auxiliary test stand. When temperature simulators are provided as auxiliary test stands, the test stands shall be portable and meet the hazardous test stand requirements of this specification. (See 4.6.2).

3.15.1 Design features. The instrumentation shall be arranged on the test panel to afford maximum visibility and operation of all instruments with a minimum amount of effort by the operator(s). Wherever practicable, instruments requiring integral lighting shall not be provided.

3.15.1.1 Instrument removal. The test panel and each instrument shall be so designed that each instrument mounted on the test panel can be readily removed from the front of the test panel or by removal of a subpanel on the front of the test panel by one man to gain access to the instrument mounting fixtures or calibration points.

3.15.1.2 Readability. Unless otherwise specified herein, all instrument scales shall be so selected that the smallest division spacing in the used portion of the sale are not less than 1/16-inch apart and shall represent not more than the specified accuracy (see 6.3.8).

3.15.1.3 Instrumentation system accuracy. Unless otherwise specified herein, the system accuracy includes the accuracy of the complete system (i.e., indicator, interconnecting wiring and cabling, transmitter, transducer, or thermocouple) (see 6.3.9).

3.15.1.3.1 Instrumentation calibration. Electrical instrumentation system calibration shall be accomplished with adjustable components in lieu of adding fixed-value components. Unless otherwise specified herein, the calibration controls shall be provided in or behind the instrument face. Zero adjustment features shall be provided on all instruments requiring calibration. All instrumentation systems shall have a minimum calibration period of 30 days.

3.15.1.4 Electrical/electrical-mechanical instruments. All electrical/electrical-mechanical instruments shall be provided with non-solder connections to permit easy installation and removal of the instruments from the test panel. These instruments shall contain no electronic tubes, except digital readout tubes. The components shall conform to current good engineering practices in regard to design, structure, strength, quality of material and workmanship.

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3.15.1.4.1 Explosion-proof or air purged (H). When electrical instruments are provided in hazardous area test stands, the indicator shall be a NFPA approved Nationally recognized laboratory listed explosion-proof for operation in a Class I, Division I, Group D or be air purged in accordance with NFPA 496, type X, entitled Purged Enclosure for Electrical Equipment, or be NFPA approved intrinsically safe for use in Class I, Division I, Group D atmosphere in accordance with NFPA 493.

3.15.2 Pressure measurement. Pressure measurement indicating systems shall be provided as required and shall provide a readout in psig unless a peculiar readout is required. All pressure indicators required for component testing shall be mounted on the test panel and the indicators not required for component testing should be mounted on the service panel.

3.15.2.1 Bourdon or diaphragm type pressure indicating systems. When bourdon or diaphragm indicating systems are provided for fluid and pneumatic systems, pressure indicators shall be equipped with AN- or MS- connectors. The interconnecting hoses shall be AN- or MS- type or commercial quality and of sufficient lengths to permit removal of the instruments from the front of their mounting panel for adjustment and calibration.

3.15.2.1.1 Pressure gages. Pressure gages shall be of the recessed type with the mounting at the front of the gage. The gage shall not extend above the surface of the test panel or service panel more than 5/8-inch.

3.15.2.1.1.1 Safety devices. The following safety devices shall be incorporated into each test and service panel mounted gage, as well as any other gage used in the test stand.

a. The gage shall be so designed to prevent rupture through the front of the gage in the event of gage failure or overpressure.

b. A blowout or relief device shall be provided on the rear of the gage that will automatically relieve the gage case pressure if subjected to one half (0.5) psi above atmospheric pressure.

c. Shatterproof glass shall be provided to cover the gage face.

3.15.2.1.2 Hydraulic dampers. Hydraulic dampers shall be provided in the pressure lines where gages are subjected to continuous surges and extreme pressure fluctuations, whenever the dampers do not affect the test results.

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3.15.2.1.3 Shutoff valves. Shutoff valves shall be provided for all pressure gages mounted on the test panel. The valves shall be located as near the gage as possible. The valve shall not be used as a hydraulic damper.

3.15.2.1.4 Automatic gage shutoff valves. Automatic gage shutoff valves shall be provided for all gages that are subject to pressure beyond their range of operation. The valves shall be of the positive action type and shall provide zero leakage when the valve is in the closed position.

3.15.2.1.5 Air bleed provisions. Where maximum accuracy is required in low pressure range gages and where the fluid head in the lines will cause a significant error in the gage reading, air bleed provisions shall be provided to free gage lines of trapped air to maintain a constant liquid head on the gage.

3.15.2.1.5.1 Visible drip indicators. Visible drip indicators shall be provided in the air bleed provisions and shall be located on the test or service panel adjacent to the item it monitors. The drain from the drip indicators shall terminate into the test stand.

3.15.2.1.6 Calibration taps. Calibration taps shall be provided for each gage mounted on the test panel to permit gage calibration without removal of the gage. The tap end shall be in accordance with MS33656, Style E, Size 4, with mating pressure cap.

3.15.2.2 Duplex/Differential pressure gage air purging. Provisions shall be provided for air purging duplex/differential pressure gages. The provisions shall consist of a 1/8-inch return line to the reservoir with a shutoff valve and drip indicator in the line.

3.15.2.3 In-line numerical readout indicating system. In-line numerical readout indicating system shall be provided when rapid or numerous pressure readings must be obtained from the indicator. The system shall consist of the pressure transmitters, interconnecting wiring, and indicator. The indicator shall be of the continuous indicating type.

3.15.2.4 Measuring section. A measuring section shall be provided in dynamic fluid line(s) where an accurate pressure indication is required. The measuring section shall be a straight stainless steel tube 5 diameters in length upstream and downstream of the pressure tap. The tubing and pressure connection shall be free of any restrictions which could affect the accuracy of the indicated pressure.

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3.15.3 Flow measurement. Flow measurement indicating systems shall be provided as required and shall provide a readout in pounds per hour unless a peculiar readout is required. Each flow measurement system shall be designed to operate with the test fluid at the required density, viscosity, and temperature limits. All flow indicators shall be mounted on the test panel. The test fluid flow being measured shall be temperature controlled within ±4°F gage of where the system was calibrated. The temperature of this flow shall be indicated on the test panel within a system accuracy of ±1°F.

3.15.3.1 In-line numerical readout indicating system. Unless otherwise specified (see 6.2.o) in-line numerical readout indication systems shall be provided for flow measurement. The flow indicator shall be of the continuous indicating type not requiring a recount from zero.

3.15.3.1.1 System accuracy. Unless otherwise specified the system accuracy of the flow measurement indicating system shall be 0.6 percent of reading or better when reading within the measuring system operating range.

3.15.3.1.2 Flow transmitter pressure drop. The pressure drop across the flow transmitter shall not exceed 8 psig.

3.15.3.1.2.1 Flow transmitter installation. The flow transmitter shall be installed in a straight section of pipe. A straight section of pipe not less than 10 diameters in length shall be provided upstream of the flow transmitter and a straight section of pipe not less than 5 diameters in length shall be provided downstream of the flow transmitter.

3.15.3.2 Variable area glass tube type indicating system. When specified (see 6.2), a variable area glass tube type indicating system shall be provided for flow measurement. When multiple glass tubes are required, the glass tubes shall be assembled into a single unit including all necessary interconnecting piping and valves.

3.15.3.2.1 Scales. The scales on the glass tubes shall have logarithmic characteristics to provide uniform percentage readability at all points on the scale. The scales shall be calibrated in pounds per hour with each scale representing not more than 1.0 percent reading. This distance between consecutive scale divisions shall not be less than 1/16 inches.

3.15.3.2.2 System accuracy. The system accuracy of the flow measurement indicating system shall be 0.5 percent of reading or better when reading within the measuring system operating range.

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3.15.3.2.3 Glass tube operating pressure. The glass tube operating pressure shall not exceed manufacturer's recommended operating pressure. The glass tube shall withstand a minimum proof pressure of 100 percent over operating pressure.

3.15.3.2.4 Pressure drop. The maximum pressure drop through the glass tube indicating system shall not exceed 8 psig.

3.15.3.2.5 Vent valve(s). A vent valve(s) shall be provided on each glass tube. A visible drip indicator shall be provided downstream of each valve(s). Return line(s) to the reservoir shall be provided from the drip indicator(s). The vent valve(s) and drip indicator(s) shall be on the test panel.

3.15.3.2.6 Blowout disc. Blowout disc(s) shall be provided as required to protect the glass tubes(s). The blowout disc(s) shall be of the diaphragm-type and shall rupture when the pressure in the glass tube(s) exceeds 85 psig. The disc(s) shall be installed in an easily accessible location and designed for quick removal and installation.

3.15.3.2.6.1 Strainer. A strainer shall be provided in the glass tube downstream of the disc to prevent the ruptured blowout disc from contaminating the system.

3.15.3.2.6.2 Visible drip indicator. A visible drip indicator shall be provided in a line teed-off the fluid line returning the fluid to the reservoir from the blowout disc. The indicator shall be downstream of the disc and mounted on the control adjacent to the glass tube(s) and shall not cause back pressure on the glass tubes.

3.15.3.2.7 Cleaning. The glass tube indicating system shall be so constructed to permit glass tube cleaning without removing the glass tube(s) from the test stand designed so that each glass tube assembly can be removed as a complete unit. The indicating system shall be so designed to not require recalibration after cleaning and reassembly. An opening shall be provided in the test stand top to permit access for cleaning the glass tubes and removing the float assembly(ies).

3.15.3.2.7.1 Floats. Floats shall be guided in the glass tubes by the side wall of the tube or through the use of a center guide wire. Construction of the float guides shall be designed to permit removal of the float and guide wire, if used, for cleaning without disassembly of the indicating unit.

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3.15.3.2.8 Shutoff valves. When shutoff valve(s) are required the valve(s) shall be mounted on the test panel beneath its respective glass tube. The flow passage diameter in the valves shall be equal to or greater than the internal diameter of the piping to which they are connected. The valves shall so operate that undue hydraulic shocks or surges shall not result from their use to change flow volume or direction. The maximum torque required to operate the valve shall not exceed 20 pound-inches of torque per inch of handle diameter. Push-pull valves requiring a maximum force of 7 pounds may be utilized.

3.15.4 Speed measurement. Speed measurement indicating systems shall be read out in RPM unless a peculiar readout is required. The accuracy of the speed indicating system shall be as required by the test, but shall not be less than 1 percent of the reading in the operating range. In-line numerical indicating systems shall be provided unless otherwise specified. All speed measurement indicating systems shall be mounted on the test panel.

3.15.5 Temperature measurement. Temperature measurement indicating system shall be provided as required and shall provide a temperature readout in °F unless a peculiar readout is required. In-line numerical indicating systems shall be provided where a temperature indication of + 1°F or better is required. All temperature measurement indicating systems shall be mounted on the test panel.

3.15.6 Timers. Timers shall be provided as required. Each time shall be self-contained and shall not require external source of frequency control. The timer(s) shall be mounted on the test panel.

3.15.7 Elapsed time meter. An elapsed time meter capable of recording 9999.9 operating hours minimum shall be provided on the service panel. The meter shall operate whenever the boost/high pressure pump are operating.

3.15.8 Timer. Whenever the component tested on a stand requires the use of a timer, the timer provided shall be of the mechanical type. Whenever the mechanical timer will not satisfy the requirements, an electric timer, NFPA approved Nationally recognized laboratory listed intrinsically safe in accordance with NFPA 493, or an electric timer housed in an explosion-proof case shall be provided. The electric timer housed in the explosion case must be time resettable without removing the case cover.

3.15.9 Hydrometer (H). A hydrometer shall be provided on all test stands utilizing MIL-F-7024 type test fluid in which the specific gravity of the test fluid has an effect upon the test results.

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3.15.9.1 Connection. The hydrometer shall be so connected into the test fluid circuit that fresh samples of the test fluid can be supplied instantly from the fluid being circulated and returned to the tank by means of inlet and outlet valves. Flow through the hydrometer shall not be indicated on the flowmeter.

3.15.9.2 Thermometer. A thermometer with an accuracy of ±0.25 percent of the range shall be provided within the hydrometer or in the fluid stream to the meter.

3.15.9.3 Accuracy and range. The accuracy of the hydrometer shall be 1.0 percent of the range and shall be calibrated at the test fluid operating temperature. The range shall not be less than 0.680 - 0.850.

3.16 Accumulator(s). When accumulator(s) are required, accumulators shall be provided in accordance with MIL-A-5498. Means shall be provided for relief or accumulator gas and hydraulic pressure.

3.16.1 Charge gas. The charge gas for accumulators shall be nitrogen or inert gas.

3.16.2 Placard. A placard or permanent tag at the fill point shall indicate the type of charge gas to be used in the maximum charge pressure.

3.16.3 Pressure gage. A pressure gage shall be provided to indicate accumulator gas charge pressure.

3.16.4 Charge valve. A gas charge valve shall be provided in accordance with MS28889 for recharging the accumulator and shall be in an easily accessible location.

3.17 Government furnished utilities. Government utilities that will be available for the operation of the test stand at the Government installation shall be as follows unless otherwise specified.

Electrical:	AC-277/480 volts, 4 wire, 3 phase, 60 cycle
Water:	Cold - 70°F Maximum @ 40 to 70 psig Hot - 180°F Maximum @ 40 to 70 psig
Water drain:	Water drain back-pressure not to exceed 4 psi.
Steam:	20 psig
Shop Air:	90 psig
CO_2 Storage Bottle(s)	

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3.18 Test components, masters or standards. Engine test components, masters or standards required for preproduction and production quantities acceptance testing shall be furnished by the supplier unless otherwise specified (see 6.2.p).

3.19 Finishes and protective coatings. Cleaning, painting, plating, and anodic films shall be in accordance with MIL-STD-808. Test stand finish colors shall be as follows:

a. Exterior surface of the test stands, except where stainless steel is specified herein, shall be semi-gloss gray number 26373 conforming to FED-STD-595.

b. Interior of the test stands shall be semi-gloss white number 27886 conforming to FED-STD-595.

c. Cadmium plating usage shall be prohibited on components of the hydraulic and pneumatic test circuits and on test stand components that can come into contact with aircraft items to be tested.

3.20 Identification of product. The test stand and any loose components of the test stand shall be marked for identification in accordance with MIL-STD-130.

3.20.1 Vendor's nameplate. The vendor's standard nameplate may be installed in addition to the US Government nameplate.

3.21 Placards. All valves, gages, outlets, connections and controls shall be identified by a name-and-number placard on the panel. The same numbers shall be shown on the flow diagram and other data furnished with the test unit. A placard or permanent tag at the fill point of accumulator(s) shall indicate the type of charge gas to be used and the maximum charge pressure. (See 3.16).

3.21.1 Warning. Warning placards or other precautionary labels as necessary shall be so located and designed that the operator cannot fail to notice them before any controls, valves or switches are operated. Warning placards shall be of laminated phenolic sheet with red surface backing to result in a red placard with white engraved letters.

3.22 Operational markings. Operational markings shall be of the permanent type.

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3.22.1 Service line markings. Steam, water, and air drain lines shall be identified by a placard at the entrance to the test stand, and shall be at the rear.

3.22.2 Direction of rotation. Directions of rotation or all rotating components shall be identified with a decal or engraved or painted arrows.

3.22.3 Orifice(s) identification. In-line orifice(s) shall be wrapped with a band of yellow tape. (See 3.11.14).

3.22.4 Reference orifice(s) identification. Reference orifice(s) shall be stamped or tagged with the calibration data. (See 3.11.13).

3.22.5 Reservoir stencil. The reservoir shall be stenciled at the field location with a type of test fluid and the reservoir capacity letters of 1/2 inch minimum height shall be used.

3.23 Workmanship. The test stand, including all parts and accessories, shall be constructed and finished in a thoroughly workmanshiplike manner in accordance with high grade manufacturing practice for this type of equipment.

3.23.1 Neatness. Particular attention shall be given to neatness or wiring, installation of tubing, hose, and fittings; thoroughness of welding, brazing, drilling, forming, riveting and painting; alignment of parts and tightness of screws and bolts; and freedom from defects, burrs, and sharp edges.

3.23.2 Cleaning. The test stand shall be completely clean and the hydraulic system shall be thoroughly flushed prior to preproduction testing and inspection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the test stand shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Quality conformance tests . . . . . See 4.5.

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4.3 Test conditions

4.3.1 Standard atmospheric conditions. Unless otherwise specified, all tests required by this exhibit shall be conducted at an atmospheric pressure of 28 to 31 inches Hg, temperatures of 65° to 95°F, and a relative humidity of 15 to 100 percent. When tests are conducted with atmospheric pressure, temperature or humidity differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Apparatus and instruments. Apparatus and instruments used in conjunction with calibration and testing specified herein shall be of laboratory precision type and shall be checked to ascertain that it has been calibrated at intervals properly spaced to assure laboratory accuracy.

4.4 Preproduction testing

4.4.1 Test sample. The first test stand of each type shall be subjected to the preproduction tests specified in 4.6.

4.4.2 Test plan. The plan for testing test stand components and calibrating test stand instrumentation and the required data sheets shall be submitted to the Government procuring activity for review 30 days before the starting preliminary testing. (See 6.2.d).

4.4.3 Test reports. Test reports shall be prepared in accordance with MIL-STD-831.

4.4.3.1 Preliminary report. A preliminary preproduction test report shall be prepared and be available at the start of the first article demonstration (6.2.g).

4.4.3.1.1 Test and correlation data. One complete set of test data obtained by testing a master(s) or standard test component(s), previously calibrated shall be contained in the preliminary report. The master(s) or standard(s) shall be tested on the first article test stand in accordance with its applicable Technical Order to obtain this data (see 4.6.24). The test results of the test stand endurance test shall also be included (see 4.6.23).

4.4.3.1.2 Calibration data. Calibration data obtained during the calibration of test stand components shall be contained in the preliminary report (see 4.6.2).

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4.4.3.1.3 Safety devices and component performance testing. All testing of safety devices and test stand components to determine they operate properly shall be described and the results of the testing included in the preliminary report (see 4.6.3).

4.4.3.1.4 Reliability, maintainability and training information. The following information on reliability, maintainability and training shall be included in the preliminary test report.

a. All test stand components or system failures and maintenance required in relation to stand operating hours during preproduction testing.

b. A description of the cause of the failure and the time required to perform the maintenance.

c. Whenever design or component changes are proposed, the engineering reasons and expected effectiveness of the changes shall be substantiated.

d. A description of the US Government's test stand operator and maintenance personnel training recommended by the supplier shall be provided (see Appendix IA, 10e).

4.4.3.1.5 Noise level data. The noise levels recorded during test stand and test component preproduction testing shall be included (see 3.3.2.1), (figures 2 and 3).

4.4.3.1.6 Foolproofness results. Foolproofness tests results and hazard operating practices to be avoided shall be described and contained in the report (see 3.7.7 and 4.6.9).

4.4.3.2 Final report. The final report shall consist of the preliminary report updated to contain the results and data of the first article demonstration testing.

4.4.3.2.1 Final report submittal. Three copies of the final report shall be submitted to the procuring activity within one month after the first article demonstration (see 6.2.h).

4.5 Quality conformance test. The quality conformance tests shall consist of the following individual tests.

4.5.1 Individual tests. Each test stand shall be subjected to the following tests as described under 4.6:

a. Examination of product	See 4.6.1
b. Calibration test	See 4.6.2

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c. Components tests . . . . . See 4.6.3, 4.6.7, 4.6.8, 4.6.12, 4.6.13, 4.6.19, and 4.6.21

d. System tests . . . . . See 4.6.25

4.6 Test methods

4.6.1 Examination of product. The test stand shall be inspected to determine compliance with the requirements specified herein. It shall be inspected for workmanship, finish, materials, markings, dimensions, meeting wiring coding, neatness of wiring; and compliance with codes and certifications as described herein.

4.6.1.1 Serviceability. The test stand shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, operation, and safety. Particular note shall be made with regard to provisions to prevent accumulation of dirt, etc., which may hinder servicing and operation. (See 3.7.3).

4.6.2 Calibration tests. The complete instrumentation and measuring systems shall be checked for calibration to determine compliance with the accuracy requirements specified in the applicable Technical Orders for the component(s) to be tested. Unless otherwise specified (see 6.2), calibration points shall be at 10, 20, 40, 50, 60, 75, 90, and 100 percent of the complete range to insure complete calibration of the instrument throughout the entire range. Each instrument or indicator and transducer or transmitter shall also be calibrated independently from the other portion of the instrument to allow exchange of major components. (See 3.15.1.3).

4.6.3 Safety devices tests. All safety devices shall be tested to demonstrate their proper operation. Where complete demonstration is impractical, a simulation or partial operation test shall be performed. The test plan (see 4.4.2 and 6.2.d) shall describe the intended test.

4.6.3.1 Noise level test. It shall be demonstrated that the noise levels produced by the test stand do not exceed the sound pressure level values specified in 3.3.2.1 in any octave band at any of the measurement positions specified in figure 1. All noise measurements shall be made during the simultaneous operation of all components of the test stand at maximum noise level. In addition, all applicable aircraft components tested on the test stand shall be installed and operated at maximum noise level. The acoustical measurement instrumentation shall be as specified in 4.6.3.1.2. The noise level tests shall be successfully completed before a first article demonstration is proposed. The noise level data shall be recorded in figures 2 and 3.

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ACOUSTICAL SURVEY DATA

Test Stand Part Nr. | Date

Test Stand Nomenclature

Test Stand Manufacturer | Engine Manufacturer

Max. Rated Capacity Data:

Description of Operation @ Test Stand Max Rated Capacity i.e. RPM, Flow & Pressure

Description of Test Component @ Max. Rated Capacity i.e. RPM, Flow & Pressure

Acoustical Measurement Eqp.

	MFG	Type
Meter		
Micro-phone		
Octave Filter		
Cal. Eqp.		

Position 10 Ft. From Test Stand	Sound Pressure Level (in DB RE 0.0002 Dynes/CM^2) Octave Band Center Freq. Hz							
	63	125	250	500	1000	2000	4000	8000
Limits (DB)	85	80	76	73	71	70	69	68
#1 Measurements								
Dev From Limits								
#2 Measurements								
Dev From Limits								
Background Noise Levels								
#3 Measurements								
Dev From Limits								
#4 Measurements								
Dev From Limits								
Background Noise Levels								
#5 Measurements								
Dev From Limits								
#6 Measurements								
Dev From Limits								
Background Noise Levels								
#7 Measurements								
Dev From Limits								
#8 Measurements								
Dev From Limits								
Background Noise Levels								

FIGURE 2

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ACOUSTICAL SURVEY DATA

Test Stand Part Nr | Date

Test Stand Nomenclature

Test Stand Manufacturer | Engine Manufacturer

Note

*Position letters are from left to right while facing the front of the stand.

Remarks

Operator's Position 2 Ft. From Test Stand	Sound Pressure Level (in DB RE 0.0002 Dynes/CM^2) Octave Band Center Freq. Hz.							
	63	125	250	500	1000	2000	4000	8000
Limits (DB)	98	91	86	83	81	80	80	80
*Pos A Measurements								
Dev From Limits								
Pos B Measurements								
Dev From Limits								
Pos C Measurements								
Dev From Limits								
Pos D Measurements								
Dev From Limits								
Pos E Measurements								
Dev From Limits								
Pos F Measurements								
Dev From Limits								

Signatures

Contractor Representative _____

Government Representative _____

FIGURE 3

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4.6.3.1.1 Measurement positions. Noise level measurements shall be made at eight positions around the perimeter of the test equipment and spaced 10 feet away, as shown in figure 1. Measurements shall also be made at several positions located two feet from the operator's control panel or at the operator's heat position when utilizing sink and sumps and spaced not more than four feet apart, as shown in figure 1. All measurements shall be made with the microphone five feet above the floor and oriented toward the test stand. Ambient background noise levels shall be at least 10 decibels less than the minimum sound pressure levels specified in table I when measured at points 2, 4, 6 and 8, as shown in figure 1.

4.6.3.1.2 Instrumentation. The basic required instrumentation shall consist of a sound level meter used in conjunction with an octave band analyzer and a system calibrator. The instrumentation shall meet the requirements specified in USA Standards S1.4 and S1.11, respectively. Up-to-date calibration and frequency response data of the entire measurement system, including microphone, shall be available to permit correction of the measurement data. The range of frequency response of the system shall be from 45 to 11,200 Hz and the response stated within ±2 db. Immediately before and after the test, an acoustical calibrator shall be used for an end-to-end system calibration from the microphone to the meter. The sound pressure level (in db ref 0.0002 dynes/cm^2) produced by the calibrator, when used in conjunction with the particular microphone in the system, shall be known within ±0.5 db.

4.6.3.2 Fire extinguishing system test (H). The fire extinguishing system shall be tested by installing a small cylinder of CO_2, approximately 25 pounds, and manually releasing the system. A temperature detector shall also be tested by installing a small cylinder of CO_2 and applying heat to the detector by use of an electric lamp. Additional temperature detectors shall be tested for proper operation without the requirement for expelling the extinquishing agent. Failure of the CO_2 to release shall be cause for rejection (see 3.3.5).

4.6.3.3 Air exhaust system test (H). Sink air exhaust systems shall be observed to determine the compliance with the dimensional and physical requirements of 3.9.11.

4.6.3.4 Grounding test (H). A grounding test shall be conducted to determine that all components of the test stand are electrostatically bonded together by assuring a continuity not to exceed 25 ohms resistance between major components of the test stand and the test stand ground (see 3.10.3).

4.6.3.5 Interlocks test(s). An interlock test(s) shall be performed to verify that motors cannot be started or stopped in such a manner as to damage the test stand (see 3.10.4.5).

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4.6.3.6 Automatic interlocks tests. Automatic interlocks test shall be performed to verify that the interlocks operate properly upon air purging, cooling water, lubrication loss, or when an over-temperature condition exists (see 3.10.4.5.1).

4.6.3.7 Emergency speed braking test. The emergency speed brake shall be tested to determine that the braking rate is 300 RPM per second or faster, and the brake is activated by the emergency switch (see 3.10.11.3.2.1).

4.6.3.8 Emergency switch test. The emergency switch(es) shall be tested to verify that upon operation all electrical, air, water, hydraulic systems shall be de-energized and the emergency brake actuated. (See 3.10.14).

4.6.3.9 Interlock switch test. The interlock on the service panel door shall be operated to verify that it turns the test stand maintenance lights off when the door is closed (see 3.10.16.1).

4.6.3.10 Reservoir pressure tests. Reservoirs shall be tested to verify that they will withstand a positive or negative 0.2 psig pressure without permanent distortion or leakage as applicable or calculations shall be provided to substantiate the strength in lieu of testing (see 3.11.2.2).

4.6.3.11 Thermo-sensor actuated valve and interlock test (H). The thermo-sensor actuated valve shall be tested by applying 160°F heat with the use of an electric heat lamp. The drain valve opening and concurrent operation of the emergency stop circuit shall be observed (see 3.11.2.7.1.1 and 3.11.2.7.1.1.1).

4.6.3.12 Overtemperature switch test. The reservoir overtemperature switch(es) shall be removed from the test stand and inserted in a liquid at 97 percent of the flash point of the fluid used in the reservoir. The actuating of the switch shall be verified by a continuity test (see 3.11.2.10).

4.6.3.13 Low fluid level cutoff switch test. The low fluid level cutoff switch test may be performed while filling the reservoir to verify that the test stand shuts down whenever the reservoir is below 1/2-full position (see 3.11.2.11).

4.6.3.14 Immersion heater well test (H). Immersion heater wells automatic overtemperature shutoff switch shall be tested to verify that it shuts the heater off at 15°F above the maximum required operating temperature and that the fluid shall not exceed 80 percent of its auto-ignition temperature. The well's low fluid level switch shall be tested to verify that it shuts the stand down whenever the fluid level is less than 6 inches over the heater (see 3.11.6.4).

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4.6.3.15 Fluid temperature controls test. Fluid temperature controls shall be tested to verify that the controls will maintain the fluid temperature below 97 percent the flash point of the test fluid (see 3.10.6.5).

4.6.3.16 In-line temperature limiting switch test (H). The in-line temperature limiting switch(es) shall be tested by revoval from the stand and inserting in a hot liquid to verify operation at 97 percent of the test fluid flash point (see 3.11.6.5.1).

4.6.3.17 Hot test fluid components test (H). Hot test fluid components shall be observed to verify that they are rated by the component manufacturer to satisfactorily operate at the required temperature and pressure. It shall be observed that all components in the hot test circuit are insulated to protect the operating and maintenance personnel from burns (see 3.11.7 and 3.2.6).

4.6.3.18 Open body tests (H). Measure the test fluid temperature from an open body test to verify the fluid temperature does not exceed 90 percent of the fluids flash point temperature (see 3.11.8).

4.6.3.19 Spray chamber electroconductive glass and bond observation (H). The spray chamber shall be observed to verify that the windows are electroconductive glass and that the glass is electrostatically bonded to the test stand (see 3.11.12.1).

4.6.3.20 Blow-open exhaust panel inspection (H). The blow-open exhaust panel shall be inspected to verify the panel is free to open without creating a hazard to personnel and that it is sized a minimum of 1 square foot of opening per 15 cubic feet of chamber area (see 3.11.12.3).

4.6.3.21 Spray chamber exhaust blower test (H). The spray chamber blower system shall be tested to verify that the chamber is free of visible vapors 30 seconds after conclusion of a test (see 3.11.12.9.1).

4.6.3.22 Filter/Separator operation test (H). The filter/separator shall be tested to verify that filtration is sufficient to prevent fluid condensation in the exhaust duct when the duct is 70°F and above (see 3.11.12.9.2).

4.6.3.23 Spray chamber blower interlock test (H). The interlock between the high pressure pump and the blower shall be operated to verify that the blower will be running whenever the high pressure pump is operated (see 3.11.12.9.3).

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4.6.3.24 Pressure vessel(s) testing. Pressure vessel(s) shall be tested (suppliers certification of testing is satisfactory) at 200 percent of its operating pressure (see 3.13.1).

4.6.3.25 Air loss automatic shutdown test. The air to the stand shall be shut off to test the operation of the automatic shutdown of the test stand (see 3.13.2.1).

4.6.3.26 Water loss automatic shutdown test. The water to the stand shall be shut off to test the operation of the automatic shutdown of the test stand (see 3.14.1).

4.6.3.27 Air purging system test (H). A test shall be performed to verify that the timing devices prevent the electrical power from being applied to the instrument until a minimum of 10 volumes of purge gas has passed through the enclosure with an enclosure pressure of at least 0.1 inch of water. A test shall be conducted to verify that the pressure or flow device cuts the electrical power off to the instrument whenever the enclosure pressure falls below 0.1 inch of water. The enclosure shall be inspected to verify that the required red warning label on opening the enclosure is properly located and contains the required statement (see 3.15.1.4.1). The test shall be performed to verify intrinsically safe circuits and components are in accordance with NFPA 493.

4.6.3.28 Pressure gage safety devices verification. The pressure gage shall be observed to verify that a gage failure will not be through the front of the gage. The gage case relief valve shall be tested to verify it will relief at one half the rupture case pressure. The manufacturer rate blow-out relief pressure shall be observed to verify gage case protection (see 3.15.2.1.1.1).

4.6.3.29 Automatic gage shutoff valve test. The automatic gage shutoff valves shall be tested by exceeding the gage pressure that it is meant to protect. The pressure shall be maintained on the valve for 15 minutes, any movement of the gage shall be cause for rejection (see 3.15.2.1.4).

4.6.3.30 Glass tube flowmeter blow-out disc. It shall be observed and verified the glass tube flowmeters are equipped with blow-out discs rated to rupture at 85 psig (see 3.15.3.2.6).

4.6.3.31 Accumulator pressure test. Accumulators shall be certified or tested at 200 percent of the operating hydraulic and pneumatic pressure (see 3.16).

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4.6.3.32 Wiring and seal inspection. Electrical wiring and seal fitting shall be inspected to determine compliance with the requirements of this specification (see 3.10.7).

4.6.4 Reliability demonstration and test. Satisfactory completion of all tests set forth in this specification demonstrated compliance with quantitative reliability requirements of this specification (see 3.7.2, 4.4.3.1.4 and Appendix IB, 10e).

4.6.5 Maintainability demonstration and test. The maintainability of the test stand will be observed during the first article demonstration. Particular attention will be given to access to strainers, filters, lubrication fittings, pumps and simular high maintenance type items (see 3.7.3 and Appendix IB, 10d).

4.6.6 Warmup time test. A warmup time test shall be performed to substantiate that the test stand will warmup for testing the engine component within 15 minutes after power is applied to the test stand (see 3.7.4.1).

4.6.7 Rigid and flexible lines tests. A statement verifying that the manufacturer has rated the rigid and flexible lines to withstand 200 percent over the maximum use pressure without leakage or permanent distortion shall satisfy the requirements stated herein (see 3.7.6.1, 3.7.6.2 and 3.7.6.2.1).

4.6.8 Maximum resistance test (H). A maximum resistance test shall be performed on the steel braid hose and end fittings assembly to determine the resistance is not greater than 100 ohms (see 3.7.6.2.1).

4.6.9 Foolproofness test. Foolproofness tests shall be conducted to substantiate whenever the test stand is improperly operated or improperly connected, a hazard will not be created. Any malfunction or hazard created by these tests shall be reported. Any hazards that cannot be designed out practically shall be reported (see 3.7.7 and 4.4.3.1.6).

4.6.10 Environmental conditions. Environmental conditions during the preproduction testing and demonstration of the test stand shall be within the range of the conditions specified in 3.8.1. The environmental conditions during testing shall be recorded. A statement verifying that the stand will satisfactorily operate in the total range of environment specified herein shall satisfy the environment requirements.

4.6.11 Forklift handling provisions. The forklift handling provisions shall be examined and measured to determine the compliance with the requirement of 3.9.2.

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4.6.12 Vibration testing. The vibration of instrumentation shall be observed during the operation of the test stand to determine if the readability of accuracy is impaired (see 3.9.9 and 3.10.6).

4.6.13 Variable drive tests

4.6.13.1 Variable speed drive control test. The variable speed drive control shall be tested to determine that the drive will not start unless the control is set for minimum speed and the control provided infinitely variable speed adjustment (see 3.10.11.2).

4.6.13.2 Variable speed regulation test. The variable speed control shall be tested to verify that the variable speed regulator controls the drive RPM within ±0.1 percent of the reading (see 3.10.11.3).

4.6.13.3 Variable speed deviation test. The variable speed control shall be tested to determine that the set speed does not vary more than 1 percent of the set speed upon a sudden load variation (see 3.10.11.3.1), nor requires more than 0.5 second to dampen out after a sudden load change.

4.6.13.4 Speed and emergency speed braking test. The speed and brakes shall be tested to determine that braking is at a rate of 300 rpm per second or better (see 3.10.11.3.2).

4.6.14 Pressurized filler filter pressure drop. The pressurized filler filter's commercial data shall be obtained to verify that its pressure drop is 10 psig or less at a flow of 10 gpm (see 3.11.2.6.2.2).

4.6.15 Heater and heater exchanger rate of heating/cooling tests. Heaters and heat exchangers shall be tested or demonstrated by calculations to determine they meet the rate of heating/cooling as specified herein (see 3.11.6.1, 3.11.6.2, and 3.11.6.3).

4.6.16 Manual valve torque and push-pull force test. Manual valves shall be tested to determine they do not exceed the maximum torque or push-pull force requirement specified herein (see 3.11.9.1.1, 3.11.9.1.4, and 3.15.3.2.8).

4.6.17 Open sump/automatic drain selection test. The open sump/automatic drain selection valve shall be tested to observe operation as specified herein (see 3.11.11.3).

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4.6.18 Blow-open exhaust panel verification (H). The blow-open exhaust panel proper operation shall be verified by calculations and reported. (See 3.11.12.3).

4.6.19 Annunicator tests. Tests shall be performed to demonstrate that each annunicator circuit operates properly. Simulation of overtemperature over-RPM etc. may be proposed in the test plan. (See 3.10.18).

4.6.20 Electromagnetic interference tests. Electromagnetic interference tests shall be conducted in accordance with MIL-STD-462 test methods CE03, RE02 and RS02 (see 3.10.17).

4.6.21 Pressure fluctuations observations. Pressure fluctuations shall be observed during calibration orifices used to determine that the requirements specified herein are met (see 3.11.13.6).

4.6.22 Preliminary performance and correlation testing. A preliminary stand test shall be performed by the supplier with the first article test stand by completely testing a master(s) or standard test unit(s) previously calibrated by the prime contractor. The master(s) or standard test unit(s) shall be tested in accordance with the components applicable Technical Order. This test shall be successfully completed and acceptable correlation obtained before a first article demonstration is proposed. The production quantities shall be subjected to the same correlation testing. (See 3.8).

4.6.22.1 Operating instructions. The operating instructions to be prepared by the supplier shall be used during preliminary and first article demonstration testing of the test stand. These operating instructions shall be detailed step-by-step procedures which shall result in the testing of the test component in accordance with its applicable Technical Order.

4.6.23 Endurance test. An endurance test of two hours duration with the first article test stand at maximum pressure, flow, RPM, temperature, amperes, volts and etc. or any combination of as applicable to each type test stand shall be performed by the supplier. The endurance test shall be successfully completed before a first article demonstration is proposed.

4.6.24 First article demonstration. The first article demonstration, at the suppliers facilities shall consist of the following:

a. Joint Government and supplier review of the hydraulic, electrical, pneumatic schematics and general configuration drawings as applicable to the type of test stand.

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b. Joint Government and supplier review of the preliminary preproduction test report, particular attention will be given to calibrated data, the safety analysis reports and test data noise level data, endurance test results, test data and correlation data.

c. Joint Government and supplier examination of the fabricated first article test stand, particular attention will be given to items specified in 4.6.1.

d. The supplier shall operate the first article test stand with Government designated representatives in attendance. Test stands on which it requires less than one day to complete the engine component(s) test(s) shall be subjected to a complete demonstration. Test stands on which it requires more than five days to completely test the engine components shall be subjected to a random check demonstration. Random checks will be designated by the Government and if it is found that these random checks do not compare to previous data, a complete test will be required.

e. The supplier shall also demonstrate to the Government that the sound level and safety tests made are validated by performing a random check. The Government representative shall designate the random checks to be made.

4.6.25 Systems tests. Systems tests shall be performed on the first article test stand (preproduction) and the production test stands. The systems tests shall consist of testing a master(s) or standard test component(s) previously tested on an acceptable test stand. The systems test shall be performed in accordance with the step-by-step procedures prepared by the contractor which shall result in the testing of the test component in accordance with its applicable Technical Order. (See 3.7.4 and 3.8).

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use. The test stands are intended for testing jet engine accessories and components thereof in depot hazardous and non-hazardous areas.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether non-hazardous test stands will be designed to operate in hazardous locations (see 1.2).

c. When the safety analysis report (3 copies) will be submitted to the procuring activity other than at the design review conference (see 3.3.1). Note: Should be specified on DD Form 1423, as applicable.

d. When the supplier shall submit the test plan for review if other than 3 months after submittal of procurement specification (see 4.4.2). Note: Should be specified on DD Form 1423, as applicable.

e. When the preliminary design review shall be conducted if other than 90 days after item Part I Specification is approved (see 3.3.4.1). Note: Should be specified on DD Form 1423, as applicable.

f. When the critical design review shall be conducted if other than a minimum of one week prior to ordering of parts and starting fabrication of the test stand (see 3.3.4.2). Note: Should be specified on DD Form 1423, as applicable.

g. When the supplier shall submit the preliminary test report if other than at the first article demonstration (see 4.4.3.1). Note: Should be specified on DD Form 1423, as applicable.

h. When the supplier shall submit the final test report if other than 3 weeks after the first article demonstration (see 4.4.3.2.1). Note: Should be specified on DD Form 1423, as applicable.

i. Whether the test stand will have a warmup period greater than 15 minutes (see 3.7.4.1).

j. Whether the test stand cabinet size may be larger than 10 feet long, 7.5 feet wide, and 6-1/2 feet high (see 3.9.1).

k. When maintenance hoisting provisions are required (see 3.9.14).

l. Whether the test stand will be connected for operation from other than 277/480 volts, four wire, 3-phase, 60-cycle power (see 3.10).

m. Location of motor starters, relays, fuses, etc. (see 3.10.4.2).

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n. Location of the test stand and variable speed motor power conversion unit (see 3.10.11).

o. Whether the test stand will be equipped with glass tube flowmeters (see 3.15.3.1 and 3.15.3.2).

p. Whether test components, masters or standards, will be furnished as GFE (see 3.18).

6.3 Definitions. For the purposes of this specification, the following definitions will apply.

6.3.1 Hazardous concentration of flammable gases and vapors. Hazardous concentrations of flammable gases and vapors are emitted from test stands utilizing MIL-F-7024, type II test fluid. Hazardous concentrations of flammable gases and vapors are also emitted from test stands utilizing MIL-H-5606 and MIL-L-7808 test fluid when operated at or above the test fluid flash point. For test stands utilizing test fluids not noted, the procuring activity will determine if the test stands are the hazardous or non-hazardous type.

6.3.2 Hazardous location. Hazardous location is a location where materials/equipment are being processed/tested/operated in which flammable gases/vapors are/or may be present in the air in quantities sufficient to produce explosive or ignitable vapor-air mixtures; or test fluid having flash point of 140°F or less is utilized; or test conditions exceed flash point of test fluids utilized. (Hazardous location, when inside a building, shall include the entire room in which located and adjacent rooms if not suitably cut off as provided in the NFPA 70; and when outside of a building, it shall be that area within 20 feet horizontally and 5 feet vertically of an accessory test device operation.)

6.3.3 Hazard level; category I - negligible conditions. Hazard level, category I, negligible condition is defined as follows:

a. Conditions such that personnel error, environment, design characteristics, procedural deficiencies, or subsystem or component failure or malfunction will not result in personnel injury or test stand damage.

b. The incorporation of safety devices and sound attenuation shall not reduce the operational effectiveness (productivity) of the test stands more than 15 percent nor increase the test stand cost more than 20 percent. Upon estimating that the reference percentage constraints will be exceeded, a cost/productivity versus safety trade-off study shall be submitted with the safety analysis report specified in 3.3.1.

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6.3.4 Mean-time-between-failure. The mean life or mean-time-between-failures is defined as the arithmetical mean (average) of the operating time between failures.

6.3.5 Downtime. Downtime is defined as the active time required to perform the necessary maintenance.

6.3.6 Prime operator's position. The prime operator's position is defined as the position the operator will have full control of the test stand, will be capable of observing all instrumentation, and visual inspection of the component under test.

6.3.7 Component test position. The component test position is the position the component under test is being operated.

6.3.8 Example of readability. If a pressure system has a range of 0 to 100 psi with a system accuracy of 2 percent of full scale which equals 2 psi, scale division on the indicator should be spaced not less than 1/16-inch apart and represent not more than 4 psi (see figure 4).

6.3.9 Accuracy. Accuracy of an instrument system means the system accuracy of the instrument and transmitter combined and working as a single unit unless otherwise specified.

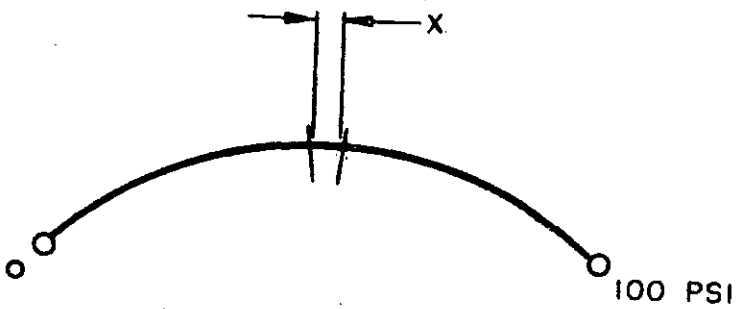
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Air Force - 11

Preparing activity:
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X = ONE SCALE DIVISION

$X \geq 1/16$ INCH

X ≤ .4 PSI

FIGURE 4. Instrument Readability

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Appendix IA

Safety Analysis Report

The safety analysis report submitted to the procuring activity shall contain the following information:

A10 Hazard analysis - A hazard analysis shall be performed as the initial analysis task during design of the test stand. This analysis shall be a comprehensive, qualitative study. Areas to be considered shall include, but are not limited to, the following:

a. Test stand environmental constraints and factors such as sound attentuation, equipment layout, lighting requirements and their safety implications.

b. Compatibility of materials.

c. Use of pressure vessels and associated plumbing, fittings, mountings, and hold-down devices.

d. Safe operation and maintenance of the system.

e. Training required for safe operation and maintenance of the test stand.

f. Fire ignition and propagation sources, and fire suppression systems.

g. Resistance to operational shock damage.

h. Fail-safe design considerations.

i. Protective clothing, equipment or devices.

j. Electrostatic protection.

k. Human error analysis of operator functions, tasks, and requirements.

A20 Operating hazard analyses - Analyses shall be performed to determine safety requirements for personnel, procedures, and equipment used in installation, maintenance, testing, transportation, operations, and training druing all phases of intended use. Engineering data,

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procedures, and instructions developed from the engineering design and initial test programs shall be used in support of this effort. Results of these analyses shall provide the basis for:

a. Design changes where feasible to eliminate hazards or provide safety devices, and safe guards.

b. The warning, caution, special inspections and emergency procedures for operation and maintenance instructions including emergency action to minimize personnel injury.

c. Identification of a hazardous period time span and actions required to preclude such hazards from occuring; and

d. Special procedures for servicing, handling and transportation.

A30 Action on identified hazards - Action to be taken to eliminate or minimize hazards revealed by analyses. If these hazards cannot be eliminated, or controlled to a specified probability of occurrence, the alternative controls will be immediately presented to the responsible procuring activity for resolution.

A40 Safety testing - Describe the tests that will be conducted to demonstrate that the safety devices operate properly.

A50 Describe all other safety aspects that are not covered under 10 through 40 above.

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Appendix IB

Design Review Data

B10 The design review data submitted to the procuring activity shall contain the following information:

a. Hydraulic, pneumatic, electrical schematics. These schematics shall also depict the connection of the test component(s) to the test stand. The hydraulic schematic shall indicate the size of the lines and fittings and the CV valve of the valves.

b. Control panel layout.

c. Parts list of major components with manufacturer data.

d. Test stand dimensions and general configuration drawing with adequate detail to allow a maintainability evaluation.

e. Reliability information on military and industrial experience on the use of major components to be used in the test stand and an engineering evaluation to substantiate why new components of the test stand should meet the reliability requirements specified (see 3.7.2). (Major components-generators-alternators).

f. Safety report.

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(H) Indicates this paragraph is applicable to hazardous test stands only.

(N) Indicates this paragraph is applicable to non-hazardous test stands only.

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